KY560351.1:9476-28410
uncultured bacterium

HTH_18 MFS_1 ? PP-binding AMP-binding_C AMP-binding Condensation AMP-binding_C AMP-binding Condensation PP-binding AMP-binding_C AMP-binding PAS His-Kinase His-Kinase Receiver BrnT_toxin ? ? ? Sigma70_r4 Sigma70_r3 Sigma70_r2 Amidohydro_1 DinB_2 ?

CP029360.1:19181-29721
Azospirillum thermophilum

MarR_2 ? HTH_ParB ParB_N HTH_26 ? Phage_integrase ? BrnA_antitoxin BrnT_toxin ? HicB_lk_antitox HTH_3 ? ? Phage_lysozyme ?

AP018207.1:3394100-3410935
Calothrix brevissima NIES-22

Methyltransf_11 Glycos_transf_2 SBP_bac_8 Peptidase_S9 Peptidase_S9_N ? URO-D Epimerase BrnT_toxin BrnA_antitoxin B12-binding Radical_SAM DUF4070 FA_hydroxylase Hydrolase HHH_5 DNA_processg_A DprA_WH His-Kinase His-Kinase PAS PAS Psb28

CCNE01000003.1:93487-103096
Mesorhizobium plurifarium

HTH_20 AHSA1 3-dmu-9_3-mt 3-dmu-9_3-mt AHSA1 TerC YmdB HTH_3 BrnT_toxin ? 5-FTHF_cyc-lig AAA Acetylase ? VWA_CoxE SelR

ADZZ01000343.1:52474-62473
delta proteobacterium NaphS2

BrnA_antitoxin DDE_Tnp_ISAZ013 HTH_ParB ParB_N IstB_IS21 Mu-transpos_C_2 HTH_38 DDE_Tnp_1 ? ? ? ? ? ? ? BrnT_toxin ?

JH603164.1:208-7784
Thiorhodovibrio frisius

REase ? ? ? ? ? Y1_Tnp BrnT_toxin ? NTP_transferase Asp_protease_2 RelE HTH_31 UPF0175 UPF0175

BOKO01000005.1:47017-56205
Chloroflexota bacterium

MFS_1 MFS_1 TetR_C_11 TetR_N Ribosomal_L21p Ribosomal_L27 Ribosomal_L31 TK Hist_deacetyl BrnT_toxin ? HTH_5 Calci_bind_CcbP Bac_luciferase ? ? SH3_3

BTRH01000523.1:323-3587
Acidobacteriota bacterium

TPP_enzyme_C BrnT_toxin ? POR ? PIN

DMEQ01000662.1:5328-7419
Cyanobacteria bacterium UBA12227

? ? Guanylate_cyc GAF FHA HTH_25 BrnT_toxin BrnA_antitoxin P-II Ham1p_like

DPEV01000467.1:731-4072
Chloroflexi bacterium

CopG_antitoxin BrnT_toxin Unstab_antitox ? DUF6933 ? BrnT_toxin ? ?

DQYS01000027.1:1-3279
Chloroflexota bacterium

SurE Peptidase_M20 ? BrnT_toxin ?

DRKM01000009.1:1-2623
Gammaproteobacteria bacterium

? BrnT_toxin NTP_transf_2 HepT-like N6_Mtase HsdM_N

DASRRM010000385.1:269-12248
Tepidisphaeraceae bacterium

DUF892 VIT VWA ABC-3 SMI1_KNR4 ABC_tran Cpn60_TCP1 BrnT_toxin DUF2442 HicB_lk_antitox HicB_lk_antitox ? ? Cpn10 Cpn60_TCP1

DASSBA010000070.1:1-5842
Gemmatimonadales bacterium

DUF1801 BrnT_toxin ParE_toxin Antitox_RHH ? DUF1428 DinB_2 Unstab_antitox ?

DASSJC010000104.1:1-3395
Burkholderiales bacterium

AMP-binding AMP-binding_C ? BrnT_toxin TctC DinB HbpS-like

DASSJS010000243.1:1-10496
Longimicrobium sp.

? DUF3108 BrnT_toxin ? Transpeptidase PBP_dimer Rho_RNA_bind ATP-synt_ab Sod_Cu Peptidase_M56 Penicillinase_R Peptidase_M56 TonB_C

DASTBD010001081.1:1-949
Candidatus Eisenbacteria bacterium

MFS_1 BrnT_toxin

DASTFQ010000078.1:1-2080
Mesorhizobium sp.

? HTH_3 BrnT_toxin rve_2 rve HTH_21 HTH_Tnp_1

DASTNO010000026.1:1-12152
Gallionella sp.

Arginosuc_synth Arginosuc_syn_C OTCace_N OTCace Aminotran_3 Ribosomal_S20p ? BrnT_toxin MurJ T5orf172 Nitroreductase HTH_1 LysR_substrate TauE NIR_SIR_ferr NIR_SIR NIR_SIR_ferr ?

DASTZM010000703.1:1-2314
Chloroflexota bacterium

? BrnA_antitoxin BrnT_toxin AP_endonuc_2 Amidase

DASYWQ010000183.1:1-2635
Candidatus Acidiferrales bacterium

? ? Peptidase_S9 ? BrnT_toxin

DASYZA010000275.1:4129-3687
Candidatus Sulfotelmatobacter sp.

BrnT_toxin CopG_antitoxin DUF5686 ?

DASZAH010000196.1:1345-8757
Steroidobacteraceae bacterium

HTH_1 LysR_substrate SnoaL_3 CcdA CcdB PEMT ANKs BrnA_antitoxin BrnT_toxin ? DUF234

DASYOV010000501.1:1-4821
Pyrinomonadaceae bacterium

Peptidase_S41 Transket_pyr Transketolase_C ExbD ? BrnT_toxin adh_short_C2 DEAD

DASYUQ010000078.1:10674-20671
Candidatus Methylomirabilota bacterium

? ? ? Peripla_BP_4 RES Xre-like-HTH Xre_MbcA_ParS_C ? BrnT_toxin ABC_sub_bind Ion_trans_2 TraG_N AhpC-TSA AhpC-TSA ABC_sub_bind APH NTP_transferase

DASYVC010000454.1:1-1661
Lysobacter sp.

DUF4386 BrnT_toxin ?

DATAMC010000280.1:3412-13931
Thermodesulfobacteriota bacterium

S-methyl_trans Acetylase DUF4892 OmpA ? AF2212-like gag-asp_proteas ? BrnT_toxin ? ZnR Ribosomal_L12 ? PIN ? YjeF_N Carb_kinase TsaE

DATARR010000163.1:101996-116945
Thermoanaerobaculia bacterium

? Sigma70_r4_2 Sigma70_r2 zf-HC2 ? RNaseH-CATRA+Caspase ? BrnA_antitoxin BrnT_toxin DUF2784 MR_MLE_N MR_MLE_C ? ABC1 ? ? Alpha-amylase

DATAZS010000332.1:1-7166
Polyangia bacterium

DEAD Helicase_C TRCF Zn_peptidase BrnT_toxin ? ? RNase_H_2 Phage_holin_4_2 SurA_N_3 Rotamase_2 SurA_N_3 Rotamase_3

DATBCN010000209.1:1-4767
Candidatus Angelobacter sp.

ABC_tran ABC_tran_Xtn ABC_tran PMSR SelR BrnT_toxin ? ? ABC-ATPase ABC_tran

DATBAQ010001470.1:1-5927
Urbifossiella sp.

ABC_tran ABC_tran BrnT_toxin ? BPD_transp_2 ? LpxB Fe_dep_repress Fe_dep_repr_C

DASUGD010000216.1:3346-14011
Gemmatimonadaceae bacterium

Esterase DUF1801 OmdA ? ? DUF6196 ? ? BrnT_toxin Peptidase_M66 DUF2306 MacB_PCD FtsX MacB_PCD FtsX PadR zf-dskA_traR ?

DUZT01000062.1:69-5605
Desulfuromonadales bacterium

ThiS BPD_transp_1 ABC_tran ABC-ATPase TOBE BrnT_toxin ? HicB_lk_antitox NTP_transf_2 HEPN ? PIN PIN

DATFGX010000037.1:19988-41247
Blastocatellia bacterium

CENP-H Polyketide_cyc ADH_zinc_N ADH_N Cyt_bd_oxida_I Cyt_bd_oxida_II nSTAND2 STY-kinase ? BrnT_toxin CarboxypepD_reg CarboxypepD_reg Reg_prop Reg_prop Reg_prop ATPase_2 PilZ DnaJ GGDEF PPDK_N Glyco_hydro_64 TPRs TPRs TPRs Cytochrome_C554

DATFHD010000071.1:1248-19667
Pyrinomonadaceae bacterium

TAT_signal Amino_oxidase Amino_oxidase DUF4440 DUF4440 PhdYeFM_antitox FMN_red ABC2_membrane_2 ABC2_membrane_4 ABC_tran ? BrnT_toxin Aconitase Aconitase_C ? DUF3298 PdaC DUF488 DUF488 TIR Ras Beta-prop_THOC3 WD40_Prp19 TIR Ras ?

DATFHD010000109.1:102432-117222
Pyrinomonadaceae bacterium

Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 ? DUF6036 FAD_binding_3 ? Oxidored_FMN Oxidored_FMN ? BrnT_toxin ? Aminotran_5 AMP-binding AMP-binding_C 4HBT_2 ? Beta_helix Cu-binding_MopE ?

DATERD010000217.1:22169-39717
Terracidiphilus sp.

DinB_2 ? PmbA_TldD_3rd PmbA_TldD_2nd PmbA_TldD_1st ? ? ? ? BrnT_toxin Glyco_tranf_2_3 ? TIG ?

DATEOX010000543.1:64-7434
Thermoanaerobaculia bacterium

Big ABC_transp_aux ? ? FA_hydroxylase ? ADH_zinc_N ADH_N BrnT_toxin ? BTAD TPRs

DATELK010000091.1:31335-50487
Pyrinomonadaceae bacterium

PKD_4 PKD_6 OmpA Sortilin-Vps10 Sortilin-Vps10 Sortilin-Vps10 PG_binding_1 Sortilin-Vps10 S1 S1 S1 S1 S1 S1 Bac_DNA_binding HIT HD 7TM-7TMR_HD 7TMR-HDED RsgA_GTPase MMR_HSR1 GTP-bdg_M GTP-bdg_N BrnT_toxin ? Vgb_lyase Peptidase_M50 tRNA-synt_1b SMC_ScpA SMC_ScpB S4 PseudoU_synth_2

DATEAQ010000296.1:1-2338
Candidatus Methylomirabilis sp.

PDT ACT Aminotran_1_2 BrnT_toxin BrnA_antitoxin

DATDRR010000341.1:1-3583
Pyrinomonadaceae bacterium

PEMT RNase_T BRCT RuvC_III MtlR ? BrnT_toxin ?

DATGDJ010000553.1:1-4479
Anaerolineales bacterium

? ? BrnT_toxin MMR_HSR1 MMR_HSR1 KH_dom-like YbbR YbbR YbbR YbbR DAC

DATHCL010000079.1:6334-15763
Xanthobacteraceae bacterium

PEP_mutase Acetylase Acetylase Epimerase SfsA_N SfsA Peptidase_M24 ? UPF0758_N RadC BrnT_toxin HTH_3 ? DUF5919 Peptidase_M15_4

DAOLBT010000006.1:31723-45538
Pyrinomonadaceae bacterium

NTP_transf_2 HepT-like Ribonuc_L-PSP Fer4_20 Pyr_redox_2 adh_short Asp_Arg_Hydrox ABG_transport ABG_transport BrnT_toxin BrnA_antitoxin ? DHO_dh Fer4_21 ? APH Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 PilM_2

DAOQZI010000002.1:242849-257369
Phycisphaerae bacterium

TPRs CsgG LPP20 LPP20 LPP20 ? T2SSE Aminotran_1_2 ? ? BrnT_toxin ADH_N ADH_zinc_N Acetylase Acetylase dCMP_cyt_deam_1 PEP-CTERM HEATs DAO ?

DAORGP010000549.1:1-2498
Candidatus Rifleibacterium sp.

MPTase-PolyVal ERCC4 BrnT_toxin ?

DAOPPX010000001.1:366521-15939
Verrucomicrobiota bacterium

Sel1 Sel1 Sel1 Sel1 Sel1 Lactamase_B_2 ? ? DNA_processg_A ? BrnT_toxin ? Herpes_UL21 ? ? DUF3375 DUF4194 SbcC_Walker_B AAA_29

DAOPNT010000009.1:44151-64107
bacterium

Glyco_hydro_2_N DUF4838 Methyltransf_25 Glyco_trans_4_4 Glyco_trans_1_4 Glyco_trans_2_3 Glycos_transf_2 TPR_CcmH_CycH Wzy_C DUF3858 Transglutaminase DUF3857 TPRs RhaB_D2 BrnA_antitoxin BrnT_toxin Hexapep Hexapep PALP ? DDE_Tnp_1_4 ? BNR_2 tRNA-synt_1g Anticodon_3 tRNA_bind

DAOSDJ010000080.1:2581-12684
Candidatus Rifleibacterium sp.

iHEPN-DUF4129 DUF4209 DUF4177 DUF3990 DUF3990 ? BrnT_toxin CopG_antitoxin REase ResIII Helicase_C EcoEI_R_C SLFN_AlbA_2 Methylase_S Methylase_S Methyltransf_11 ? ?

DAOOTQ010000001.1:37255-51679
Candidatus Competibacteraceae bacterium

Adenylsucc_synt ? NrtR_WHD NUDIX HlyD_D23 UPF0150 ? ? BrnT_toxin ? Bro-N HTH_38 FtsX MacB_PCD ABC_tran TrkA_N TrkA_C TrkH CitMHS

DAOPIE010000030.1:26719-45199
bacterium

Sigma70_r4_2 Sigma70_r2 HEATs ? A9CJY8-like_N ACT_7 ? Cellulase DUF4460 BrnT_toxin AP-GTPase COR-A COR-B Polysacc_lyase UPF0158 GH97_N Glyco_hydro_97 GH97_C UxuA UxuA CelD_N Glyco_hydro_9 CopG_antitoxin

DAOOXP010000222.1:611-9454
Candidatus Anammoximicrobium sp.

Esterase BrnA_antitoxin BrnT_toxin Toprim Topoisom_bac Toprim_Crpt URO-D Dabb STY-kinase Sulfatase

DAORWQ010000271.1:67-3687
Anaerolineales bacterium

BrnT_toxin BrnA_antitoxin DUF2157 ? ? ABC-ATPase ABC_tran

DAOMSB010000182.1:1-4025
Promineifilum sp.

? Bactofilin HD ? BrnT_toxin PGK

DATIFQ010000413.1:42-4973
Bryobacteraceae bacterium

Amidohydro_2 ? BrnT_toxin 3-dmu-9_3-mt ? DUF2442 DUF4160 Glyoxalase HTH_37 Gp49

DATIFO010000140.1:41599-60160
Candidatus Methylomirabilis sp.

DUF4331 URO-D SBP_bac_5 DoxD HD_5 HAMP NDH2_C Pyr_redox_2 DUF2284 BrnT_toxin TPRs RMD1-3 NAD_binding_4 Glycos_transf_2 GDE_C GDE_N YiiM_3-alpha MOSC Glyoxalase Glyoxalase Pyr_redox_dim Pyr_redox_2

DATIBY010000181.1:3480-2915
Blastocatellia bacterium

TIR PapC_N ? BrnT_toxin

DATIOQ010000049.1:1-2025
Rhodanobacteraceae bacterium

BrnT_toxin BrnA_antitoxin TsaD GST_C GST_N_2

DATILU010000045.1:1-2313
Burkholderiales bacterium

? BrnT_toxin ? DUF4258 DUF2130

DATHNK010000375.1:1-8950
Bacteroidia bacterium

OprB ? DUF1801 ? BrnT_toxin PIG-L GCV_T_C GCV_T 2-ph_phosp S1-P1_nuclease MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2

DATHEF010000039.1:63360-78747
Thermoanaerobaculia bacterium

? ABC_tran Pyridoxal_deC FKBP_N FKBP_C YkuD PG_binding_1 Scaffold AceK_regulatory AceK_kinase ? BrnT_toxin Kazal_1 CSD Calcineurin ? GMC_oxred_N GMC_oxred_C Ring_hydroxyl_A Rieske Histone_HNS_N

DATJGK010000027.1:1016-6398
Gemmatimonadaceae bacterium

Tannase Cu-oxidase_3 ? Adhesin_E HTH_Tnp_1 HTH_21 rve rve_3 ? ? BrnT_toxin ? HKD

DATJVR010000072.1:15277-8655
Xanthobacteraceae bacterium

Ribonuc_red_lgC Ribonuc_red_lgC TSCPD BrnT_toxin HTH_3 dCMP_cyt_deam_1 ? GIY-YIG tRNA-synt_1f Zn_ribbon_IS1595 ?

DATKZS010000050.1:19325-30552
bacterium

POR_N PFOR_II Fer4_21 PhosphMutase YfhO YfhO DUF2281 ? ? BrnT_toxin PALP NAD_binding_3 Homoserine_dh ACT Aminotran_1_2 ?

DATLUO010000011.1:6890-28582
Pirellulales bacterium

Malt_amylase_C Alpha-amylase OEP OEP HlyD_D23 ACR_tran N6_Mtase N6_Mtase DNA_processg_A ? N6_Mtase REase BrnT_toxin ? DUF3102 Methyltransf_25 TPRs TPRs ARM_TT21_C YfiO ARM_TT21_C TPRs TPRs TPRs TPRs TPRs MotA_ExbB ExbD Trypsin PDZ_6

DATLTH010000394.1:1-2191
Stellaceae bacterium

Bac_luciferase BrnA_antitoxin BrnT_toxin Lactamase_B B_lactamase-like_C

DATLMD010000170.1:213-4520

Longimicrobium sp.

CarboxypepD_reg Plug CarboxypepD_reg Plug ? BrnT_toxin PMSR Methyltransf_31

DATKND010000019.1:3774-16813
Myxococcota bacterium

PEMT Phage_holin_4_2 GST_N_2 GST_C_2 Ribonuc_L-PSP Metallophos_2 DUF4349 ? BrnT_toxin ? SPASM Usp Usp Peptidase_S8 PEP-CTERM YkuD YkuD

DATKMS010000308.1:1891-11294
Candidatus Margulisiibacteriota bacterium

ParE_toxin Unstab_antitox Y2_Tnp Phage_int_SAM_4 Phage_integrase Transposase_20 DEDD_Tnp_IS110 BrnA_antitoxin BrnT_toxin ? DUF433 ? ParE_toxin ParD_antitoxin LptF_LptG LptF_LptG

DATKKP010000140.1:27788-46425
Candidatus Eisenbacteria bacterium

Glyco_hydro_76 Thioredox_DsbH Fer4_8 CCG CCG FAD_binding_4 FAD_binding_4 FAD-oxidase_C CBS CBS adh_short_C2 HTH_3 BrnT_toxin TsaE YjeF_N Carb_kinase Ion_trans_2 TrkA_N TrkA_C AA_permease_2 AA_permease_2 TMP-TENI Biotin_carb_N CPSase_L_D2 Biotin_carb_C

DATKCU010000236.1:1-2100
Planctomycetota bacterium

? BrnT_toxin HTH_3 PQQ_2

DASZOZ010000007.1:20018-32671
Stellaceae bacterium

Cytochrom_C Cytochrom_C DUF3734 Patatin BKACE CoA_trans CoA_trans DUF2721 Glyoxalase BrnT_toxin Lactamase_B B_lactamase-like_C G_glu_transpept UreD Urease_gamma Urease_beta Amidohydro_1 Urease_alpha UreE_C UreE_N

DATMET010000011.1:1-2480

Longimicrobium sp.

G_glu_transpept BrnT_toxin ?

DATMVC010000301.1:1-5284
Ktedonobacteraceae bacterium

PBP_dimer_2 Transpeptidase Chorismate_synt HTH_3 BrnT_toxin P22_AR_N DUF3800

DATMRP010000621.1:1-2874
Thermoanaerobaculia bacterium

AA_kinase BrnT_toxin GBBH-like_N PseudoU_synth_1 PseudoU_synth_1 AIRS_C AIRS

DATFFW010000308.1:478-1321
Candidatus Paceibacterota bacterium

? BrnT_toxin

DATNLO010000172.1:30-5705
Candidatus Binataceae bacterium

New_glue GT87 FtsX MacB_PCD ABC_tran HlyD_D23 BrnT_toxin

DATNHE010000659.1:1-3012
Candidatus Binataceae bacterium

Bac_luciferase BrnA_antitoxin BrnT_toxin RsgA_GTPase MMR_HSR1 GTP1_OBG AA_kinase

DASPZC010000434.1:8135-20096
Xanthobacteraceae bacterium

Peptidase_S11 ? TAT_signal SBP_bac_5 BPD_transp_1_N BPD_transp_1 OppC_N BPD_transp_1 ABC_tran oligo_HPY ABC_tran oligo_HPY BrnT_toxin HTH_3 Amidohydro_3 Amidohydro_3 HicB_lk_antitox MR_MLE_C MR_MLE_N UreF UreE_C

DASPXX010000268.1:28135-12281
Thermoanaerobaculia bacterium

DNA_topoisoIV CopG_antitoxin DUF5677 BrnT_toxin His-Kinase DNA_gyraseB Toprim GyrB_hook GyrB_insert DNA_gyraseB_C ? SMC_N DNA_pol3_beta DNA_pol3_beta_2 DNA_pol3_beta_3 DnaA_N Bac_DnaA Hda_lid Bac_DnaA_C UDPGT Glyco_trans_4_4 ?

DASQZV010000097.1:1-9661
Terriglobales bacterium

Pribosyltran ATP_bind_3 TilS TilS_C Plug CarboxypepD_reg OmpH Ala_racemase_C Ala_racemase_N BrnA_antitoxin BrnT_toxin ABHD18

WOUQ01000017.1:25269-38449
bacterium

? Na_sulph_symp CBS CBS Receiver MFS_1 Methyltransf_32 TOBE_3 DMT_6 BrnT_toxin ? NYN OST-HTH Lactamase_B HNH SHMT Zn_ribbon_NrdR ATP-cone dCMP_cyt_deam_1 RibD_C

JSZA02000069.1:3913-22292

Candidatus Thiomargarita nelsonii

DcaP DAHP_synth_1 ? ? VapB_antitoxin DUF4153 YtxH ? BrnT_toxin Enolase_N Enolase_C Cytochrom_C_2 ? ? TMEM43 FGS STY-kinase

LGRD01000001.1:25525-39812
Methylibium sp. NZG

FAD_binding_5 CO_deh_flav_C ABC_tran BCA_ABC_TP_C BPD_transp_2 BPD_transp_2 ? ? Unstab_antitox BrnA_antitoxin BrnT_toxin ? Peripla_BP_6 AA_synth AA_synth Fer2 Fer2_2 Ald_Xan_dh_C MoCoBD_1 MoCoBD_2 ABC_tran BrnT_toxin

LDXO01000001.1:257486-270472
Deltaproteobacteria bacterium CSP1-8

CBM_48 Alpha-amylase Alpha-amylase Alpha-amylase_C ? adh_short Esterase ATP-grasp_4 CPSase_L_D2 DUF7918 DUF4440 BrnT_toxin BrnA_antitoxin Methyltransf_11 Patatin DLH DUF4154 CHASE8 HAMP His-Kinase His-Kinase

PACL01000040.1:5545-23202
Fulvimarina sp.

GCV_T Beta-barrel_CAF17_C Tc-R-P Guanylate_cyc Lactamase_B_2 cNMPBD ABC_tran ABC_tran ABC_membrane oligo_HPY ABC_tran oligo_HPY ABC_tran BrnT_toxin BrnA_antitoxin Glyco_tran_28_C Glyco_trans_1_4 Glyco_tran_28_C Guanylate_cyc Acetylase ?

JACCRE010000236.1:364-10341
Planctomycetaceae bacterium

SASA MatE MatE HTH_17 Receiver BrnT_toxin BrnT_toxin BrnA_antitoxin ? BrnT_toxin ? PMT_2 ? MoaC

PAQT01000077.1:272-10272
Lysobacterales bacterium

Arabinose_bd HTH_18 GBBH-like_N Ubie_methyltran SCP2 BrnT_toxin BrnA_antitoxin DUF7668 ? cNMPBD EAL Beta-lactamase ? ?

VJOX01003038.1:85-7821
Nostoc sp. HG1

GATase ? BrnT_toxin BrnA_antitoxin ? ? ? Nucleos_tra2_N Gate Nucleos_tra2_C DUF2079 Nucleos_tra2_N Gate Nucleos_tra2_C

JACIYZ010000005.1:234711-253519
Rhizobium sp.

MoCF_biosynth FLAD1_M Usp Pribosyltran Beta-Roll Beta-Roll Peptidase_M10_C Reprolysin_5 ? Ribonuc_red_2_N Ribonuc_red_lgC Ribonuc_red_lgC TSCPD Peptidase_S26 BrnT_toxin HTH_3 OMP_b-brl OsmC Acetylase Amino_oxidase GGDEF CHASE4 DUF1275

JACMOJ010000119.1:1-4067
Burkholderiales bacterium

ThiC_Rad_SAM BrnT_toxin ? HicB_lk_antitox DUF4160 DUF2442 DUF1330 Beta-lactamase Thiolase_C

JACNJN010000063.1:7445-15746
Candidatus Desulfolinea nitratireducens

CCG CCG Fer4_8 MFS_1 MFS_1_like CdAMP_rec CdAMP_rec MR_MLE_N MR_MLE_C ? BrnT_toxin ?

JACVRI010000467.1:274-2939
Coleofasciculus sp. C3-bin4

? Abi_alpha BrnT_toxin ? Ap4A_phos_N ATP_transf

JADEXX010000285.1:1-7943
Chroococcidiopsidales cyanobacterium LEGE 13417

N_methyl Pilin_GH Methyltransf_31 ? HepT-like NTP_transf_2 ? ? BrnT_toxin ? SNF2-rel_dom Intein_splicing SNF2-rel_dom DUF3670

JADEXJ010000009.1:35086-8998
Pleurocapsales cyanobacterium LEGE 10410

? CbiA GerE ? Sipho_Gp157 BrnT_toxin BrnA_antitoxin DUF6151 ? ? DndE Prim-Pol PriCT_2 ArdcN MPTase-PolyVal

JADGBR010000035.1:2144-19467
Candidatus Omnitrophota bacterium

HAP1_N CBS CBS Receiver His-Kinase His-Kinase Receiver OEP OEP HlyD_D23 HlyD_D23 ACR_tran OprB BrnT_toxin BrnA_antitoxin Rrf2 PBP_like_2 BPD_transp_1 BPD_transp_1 ABC_tran BrnT_toxin

JADFYK010000075.1:10112-9821
Desulfovibrionaceae bacterium

Memo BrnA_antitoxin BrnT_toxin Amidohydro_1 OTCace_N OTCace ABC_tran SUFBD_core GyrI-like APS-reductase_C Fer4_7 FAD_binding_2

JACYMF010000088.1:58433-10019

Cyanobacterium sp. T60_A2020_053

BrnT_toxin ? HicA_toxin BrnA_antitoxin BrnT_toxin BrnA_antitoxin BrnT_toxin Cons_hypoth95 ? N6_Mtase S-methyl_trans Pterin_bind B12-binding_2 B12-binding Met_synt_B12 BrnT_toxin

JACCQO010000003.1:135283-145457
Desulfovibrionaceae bacterium

? ? OmpA FecR rve_3 rve HTH_21 HTH_Tnp_1 ? Phage_integrase BrnT_toxin BrnA_antitoxin HTH_3 ? ParE_toxin HTH_3 ? Gly-zipper_Omp Gly-zipper_YMGG

JABMDF010000059.1:29544-37948
Nitrospira sp.

HTH_3 RelE Caroten_synth ? ? zf_VAL1_N ? BrnT_toxin ParE-like_toxin PhdYeFM_antitox PI3K_P85_iSH2 DUF4926

WGMT01000033.1:69580-90255
bacterium

SelA SBP_bac_1 BPD_transp_1 BPD_transp_1 Amidohydro_1 Amidohydro_1 ? DUF1485 MlrC_C BrnT_toxin ? Enolase_C Enolase_N APH eCIS_core eCIS_core ? Abhydrolase_1

JACPPP010000037.1:181-7664
Armatimonadetes bacterium

? ? ? BrnT_toxin LbH_EIF2B NTP_transferase dTDP_sugar_isom SDH_protease GDP_Man_Dehyd NLPC_P60 SH3_3 SH3_16 AP_endonuc_2 NTP_transf_5

JACPRB010000202.1:1-6959
Planctomycetota bacterium

PAS PAS GAF_2 HrpB_C HA2_C HA2_N Helicase_C DEAD Anti-TRAP BrnA_antitoxin BrnT_toxin ?

JACPWC010000061.1:707-15216
Chloroflexi bacterium

ICMT ? Cys_Met_Meta_PP PCYCGC HYDIN_VesB_CFA65-like_Ig Trp_oprn_chp DUF5915 Anticodon_1 tRNA-synt_1 ? BrnT_toxin DUF5915 TPRs TPRs AAA_16 BTAD TPR_MalT OsmC Methyltransf_31 Methyltransf_11 Peptidase_A8

JACQBY010000115.1:2086-22866
Nitrospirae bacterium

SMC_N OB_RNB CSD2 RNB S1 ACP_syn_III ACP_syn_III_C HTH_3 HigB-like_toxin ? BrnA_antitoxin BrnT_toxin HKD HKD UvrD-helicase UvrD_C PDDEXK_1 ADDB_N UvrD_C PDDEXK_1 Rrf2 ? NHL NHL NHL Cu-oxidase_3

JACQDB010000017.1:7723-19338
Acidobacteria bacterium

FAD_binding_4 FAD-oxidase_C FHA DUF309 FAD_binding_4 FAD-oxidase_C PIN PIN ? ? BrnT_toxin ThiS ThiF Rhodanese Prok-JAB PALP ThiS NinF PALP ? CBS CBS

JACQDB010000125.1:28042-39776
Acidobacteria bacterium

PilZ COXG FAD_binding_5 CO_deh_flav_C HicA_toxin HicB_lk_antitox PSII_BNR PSII_BNR Sortilin-Vps10 ? BrnT_toxin Ald_Xan_dh_C MoCoBD_1 MoCoBD_2 23S_rRNA_IVP Fer2 Fer2_2 PTPS_related ? ? MazG MazG

JACQDI010000499.1:3640-11748
Acidobacteria bacterium

Nramp MgtE_N CBS CarboxypepD_reg Plug OMP_b-brl_4 SduA_C SduA_C ? ? BrnT_toxin

JACQFG010000267.1:450-4450
Planctomycetota bacterium

ATPgrasp_YheCD BrnT_toxin Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 ACAD9-ACADV_C TPRs DUF3299 ?

JACQGR010000315.1:426-9556
Chloroflexi bacterium

? BrnT_toxin HUTI_composite_bact Amidohydro_1 ? ? ? ? ? Rieske

JACQTO010000108.1:78450-85233
Betaproteobacteria bacterium

FAD_syn Flavokinase DUF433 ? ? TIR ? ? BrnT_toxin ? ? PBP

JACQVD010000120.1:3443-8421
Gemmatimonadetes bacterium

CBM9_1 HEATs HEATs BrnT_toxin ? KGG RimK Pro_CA AICARFT_IMPCHas MGS Formyl_trans_N Hydrolase

JACQVW010000172.1:1-13725
Planctomycetota bacterium

FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 Dockerin_1 ? ? ? BrnT_toxin ? ? Hexokinase_2 ABC_tran ABC2_membrane_2 ? ABC2_membrane_2

JACQVW010000484.1:92-10060
Planctomycetota bacterium

? AA_permease_2 Methyltransf_3 BrnT_toxin BrnA_antitoxin Met_synt_B12 URO-D URO-D B12-binding B12-binding_2 ? ABC_tran

JACQYF010000210.1:7925-20732
Chloroflexi bacterium

Methyltransf_11 ABATE zf-CGNR DUF1706 ? VWA VWA_2 BatA Polyketide_cyc2 BrnA_antitoxin BrnT_toxin ? Rhomboid PD40 PD40 PD40 MacB_PCD FtsX ABC_tran Biotin_lipoyl_2 HlyD_D23 Cytochrom_C

JACQZT010000221.1:1789-10496
Acidobacteria bacterium

HTH_1 LysR_substrate peroxidase peroxidase Creatininase Sigma54_DBD Sigma54_CBD Sigma54_AID Ribosom_S30AE_C Ribosomal_S30AE PapZ_C RapZ-like_N CopG_antitoxin BrnT_toxin ?

JACRMS010000033.1:140422-150598
Nitrospirae bacterium

DUF3417 Phosphorylase ATPaseInh_sub_z ? ? TauE Usp Usp ? BrnT_toxin DUF5666 ? ArlS_N HAMP His-Kinase His-Kinase Receiver Trans_reg_C DUF6520

JACRHL010000010.1:29770-43213
Nitrospirae bacterium

Suc_Fer-like Pps Pps ? PBP_like_2 ? MS_channel_3rd MS_channel_2nd MS_channel_1st MS_channel_1st_1 BrnA_antitoxin BrnT_toxin Methyltransf_25 Patatin Pps Pps Pps TPR_MalT TPRs TPRs Peptidase_M48 Peptidase_M48 PDZ_6 ? ?

JACREE010000015.1:10441-21162
Gammaproteobacteria bacterium

IF3_N IF3_C Ribosomal_L35p Ribosomal_L20 Bac_DNA_binding SIS_2 Flavoprotein DFP FAD_syn Flavokinase BrnT_toxin BrnA_antitoxin tRNA-synt_1 Anticodon_1 zf-FPG_IleRS THDPS_N_2 Hexapep Hexapep_2 Peptidase_M20 DeoRC Rib_5-P_isom_A ?

JACRBL010000045.1:16637-31021
Chloroflexi bacterium

SBP_bac_8 VWA 23S_rRNA_IVP FGS DinB_2 Thiolase_N Thiolase_C Peptidase_MA_2 DNA_alkylation BrnT_toxin DUF3048 SdrD_B DUF3048_C DUF4062 HIT Exonuc_VII_L tRNA_anti_2 Exonuc_VII_S DUF212 ?

JADJLH010000003.1:38927-50037
Saprospirales bacterium

Lipocalin_4 CHU_C Lectin_L-type_dom ArsC MFS_1 Receiver EAD11 Glyoxalase ? BrnT_toxin ? 2-oxoacid_dh 2EXR 2-oxoacid_dh BrnT_toxin ? ? NOV_C ?

JADJIZ010000017.1:62017-80196
Gemmatimonadetes bacterium

RNaseH-CATRA+Caspase PadR MacB_PCD FtsX MacB_PCD FtsX HTH_18 DUF6597 DUF6265 ? ? BrnT_toxin BrnA_antitoxin Uma2 Uma2 SEC-C ? SLFN-g3_helicase

JADJQV010000009.1:180551-189732
Bdellovibrionales bacterium

? ? ? ? GrpE ? BrnA_antitoxin BrnT_toxin ? Fic ? Glycos_transf_2 Glyco_trans_2_3 Glycos_transf_2 Polysacc_synt_3

JADKBZ010000005.1:205913-6221

Holophagales bacterium

UvrD-helicase ? BrnT_toxin HTH_IclR UvrD_C_2 ? ? Virulence_RhuM ? ABM

JADKAE010000009.1:172392-189927
Saprospiraceae bacterium

? Plug CarbopepD_reg_2 CBM_4_9 PKD PKD DapB_N DapB_C Synaptojanin EamA EamA Peptidase_S46 BrnT_toxin ? Sortilin-Vps10 Sortilin-Vps10 PKD_4 LamininG Por_Secre_tail DUF4340 Ham1p_like ?

JADJWS010000005.1:94805-109317
Candidatus Defluviilinea gracilis

Glyco_trans_1_4 Glyco_transf_4 CBM9_1 Glyco_hydro_18 4HBT_2 Receiver GAF_2 GAF HD_5 ARG ? BrnT_toxin VWA_CoxE AAA ? ? Glyco_transf_4 Glyco_trans_1_4 DUF2180 Radical_SAM SPASM

NVRQ02000081.1:1-6514
Gammaproteobacteria bacterium

Phage_integrase BrnT_toxin BrnA_antitoxin ? ? ? ? HTH_3 Zn_ribbon_DnaG DNAG_N Toprim_2 AAA_29

JAEUQN010000067.1:19043-34349
Blastocatellia bacterium

PQQ_2 TPRs TPRs TPRs TPRs CarboxypepD_reg Plug OMP_b-brl_4 HTH_IclR Peripla_BP_4 BD-FAE Peptidase_M20 ? BrnT_toxin Methyltransf_31 AA_permease_2 CCP_MauG

VGNE01000031.1:173-7419
Chloroflexi bacterium

GTP_EFTU DEAD Helicase_C BrnT_toxin ? CSD Epimerase Pur_DNA_glyco adh_short_C2 Bactofilin tRNA_edit HGTP_anticodon

VGLH01000051.1:15606-28148
Candidatus Poribacteria bacterium

FliG_C FliG_M FliG_N YscJ_FliF_C YscJ_FliF FliE His-Kinase His-Kinase PAS FGGY_N FGGY_C HAMP His-Kinase His-Kinase Receiver NtrC-AAA AAA_lid_14 HTH_8 BrnT_toxin HTH_3 FlgD_ig DUF4096 HTH_31

VGJU01000007.1:18931-13244
Candidatus Zixiibacteriota bacterium

TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs RecA RecA_C ? BrnT_toxin ? HTH_3 Peptidase_M23 Peptidase_M19 Bac_luciferase MR_MLE_C DHDPS

VGIQ01000121.1:5952-7438
Calditrichaeota bacterium

BrnT_toxin SDH_protease TRF DUF6567 Methyltransf_12

VGYK01002655.1:1-1331
Planctomycetota bacterium

HEATs BrnT_toxin ? ?

VGYD01000483.1:357-2845
Planctomycetota bacterium

GFO_IDH_MocA_C3 GFO_IDH_MocA ? ? BrnT_toxin HEATs

VGYD01001858.1:1131-1620
Planctomycetota bacterium

? BrnT_toxin

VGSJ01000017.1:25970-41458
Deltaproteobacteria bacterium

? HNH_3 ? HTH_17 Phage_integrase Phage_int_SAM_3 Phage_integrase DUF7683 T1R_D3-like SWI2_SNF2 REase BrnT_toxin BrnA_antitoxin PDDEXK_3 Methylase_S Methylase_S N6_Mtase HsdM_N HTH_3 ? ?

VGOT01000154.1:6536-10448
Candidatus Rokuibacteriota bacterium

DUF6036 HTH_31 AAA_12 AAA_30 RNase_H_2 PDDEXK_1 BrnT_toxin ? ? BrnT_toxin G_glu_transpept DUF2442 Rhomboid DUF2459

JAFGOZ010000610.1:1-3947
Bacteroidales bacterium

His-Kinase PRiA4_ORF3 BrnA_antitoxin BrnT_toxin ParE_toxin PhdYeFM_antitox MqsA_antitoxin DUF4258

JAFGFN010000154.1:8787-11838
Candidatus Sumerlaeota bacterium

B12-binding Radical_SAM Glyco_transf_4 Glycos_transf_1 Glyco_transf_4 Glycos_transf_1 Radical_SAM SPASM ChlI Mg_chelatase Mg_chelatase_C ? DUF6036 BrnT_toxin ? GFA ? DinB_2 ?

JAFGDN010000013.1:19956-32125
Sedimentisphaerales bacterium

EamA EamA Plug TonB_dep_Rec_b-barrel DUF4154 YmdB HD KH_1 RNase_Y_N Radical_SAM HemN_C GTP_EFTU GTP_EFTU_D2 EFG_C LepA_C BrnT_toxin ? ? ?

JAFGRB010000434.1:3309-23204
Deltaproteobacteria bacterium

DcrB fn3 fn3 fn3 fn3 TIG TIG DUF3280 PEGA ? Adenylsucc_synt Castor_Poll_mid TrkA_N Ion_trans BrnA_antitoxin BrnT_toxin ? CarboxypepD_reg Plug TonB_dep_Rec_b-barrel Ribonuc_L-PSP ? CN_hydrolase Kelch_KLHDC2_KLHL20_DRC7 PEGA

JAFGWI010000133.1:146-26541
Spirochaetales bacterium

? BrnT_toxin ? DEDD_Tnp_IS110 Transposase_20 ? ? Glucodextran_B FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 fn3 ? Glutaredoxin

JAFGTX010000417.1:295-1087
Clostridia bacterium

HicA_toxin ? BrnT_toxin

JAFLCN010000014.1:12900-28192
Planctomycetota bacterium

ECH_1 ECH_1 Zn_Ribbon_1 ECH_1 Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 ACAD9-ACADV_C Abhydrolase_6 E1_dh BrnT_toxin ? Transket_pyr Transketolase_C Biotin_lipoyl E3_binding 2-oxoacid_dh Y1_Tnp LacI Peripla_BP_3 ? ?

JAFZZR010000099.1:33106-50897
Lentisphaeria bacterium

Glyco_hydro_20 Glyco_hydro_20b Glyco_hydro_10 Glyco_hydro_9 CelD_N CBM_4_9 SBP_bac_10 N_methyl Glyco_hydro_2_C Glyco_hydro_2_N2 LacI Peripla_BP_3 Melibiase_C Melibiase_2 Melibiase_2 He_PIG BrnT_toxin BrnA_antitoxin VWA_2 BatA iHEPN-DUF4381 vWA AAA_lid_2 AAA_3 His_Phos_1 ?

JAFZXT010000168.1:1-4662
Neisseriaceae bacterium

Amidase PfkB Porin_1 BrnT_toxin BrnA_antitoxin ? CRISPR_Cas2

JAGHZG010000181.1:1-4897
Chloroflexota bacterium

GDP_Man_Dehyd RHH_5 BrnT_toxin ? ATP_bind_3 ATP_bind_3 UPF0175 DUF3368 SMC_N SMC_hinge

JAGNRC010000065.1:24356-11251
Chloroflexota bacterium

HupF_HypC ? BrnT_toxin Acylphosphatase zf-HYPF zf-HYPF Sua5_yciO_yrdC HypF_C HypF_C_2 SH3_3 cNMPBD SpoIIE HHH_8 HHH_5 DNA_pol_B_palm DNA_pol_B_thumb PHP SBP_bac_5 SBP_bac_5 BPD_transp_1 BPD_transp_1_N BPD_transp_1 OppC_N

JAFNZS010000104.1:1-5575
Oscillospiraceae bacterium

ABC_tran BrnA_antitoxin BrnT_toxin Beta_elim_lyase Peptidase_S41 Alpha-amylase

JAFRMU010000124.1:106-5360
Butyrivibrio sp.

HTH_28 PhoU ? DUF3847 MobA_MobL BrnT_toxin BrnA_antitoxin ?

JAFZHS010000019.1:165214-183047
Clostridiales bacterium

? HTH_3 OKR_DC_1 OKR_DC_1_C Spermine_synt_N Spermine_synth Arginase Sacchrp_dh_NADP Sacchrp_dh_C Orn_DAP_Arg_deC BrnT_toxin CopG_antitoxin Zn_Ribbon_TF ABC_tran ABC2_membrane_2 Asp_Glu_race ? CBAH

JAFYHO010000079.1:246-27232
Actinomycetaceae bacterium

tRNA-synt_1b ? HTH_26 DUF6290 ? BrnT_toxin CopG_antitoxin RelB YoeB_toxin CdiA_C SpaA Cna_B FctA FctA FctA Cna_B Cna_B Cna_B Cna_B Cna_B Cna_B Cna_B Cna_B Cna_B Cna_B Cna_B CshA_repeat Muc_B2 SpaA Flg_new Flg_new Flg_new Flg_new Flg_new Malt_amylase_C Alpha-amylase Alpha-amylase CBM_48 NUDIX-like Zn_ribbon_NUD NUDIX

JAFWSO010000072.1:1-2257
Ottowia sp.

Porin_4 BrnA_antitoxin BrnT_toxin Porin_4

JAFEDF010000138.1:272-1408
Proteobacteria bacterium

BrnA_antitoxin BrnT_toxin ParE_toxin ?

JAANWX010000077.1:22839-39033
Chloroflexota bacterium

Glycos_transf_1 Glyco_transf_4 PIG-L CC_BshC Rossmann-like_BshC Methyltransf_25 Lactamase_B Lactamase_B ClpB_D2-small AAA_2 AAA_lid_9 AAA Clp_N Clp_N S1 S1 S1 S1 BrnT_toxin BrnA_antitoxin AAA_16 Mg_chelatase AAA_lid_2 VWA AAA_23 GAS Rad50_zn_hook SbcC_Walker_B

JAGXSO010000089.1:8094-16004
Truepera sp.

PIN VRR_NUC UDG ? HicA_toxin HicB_lk_antitox BrnA_antitoxin BrnT_toxin ? Aminotran_3 PBP_like_2 BPD_transp_1 ABC_tran

JAHJAI010000074.1:58368-70946
Gammaproteobacteria bacterium

Glyco_trans_2_3 Ribosomal_L13 Ribosomal_S9 RelB ? Porin_4 BrnA_antitoxin BrnT_toxin ? ? ? PemK_toxin DUF29 DUF3683 FAD_binding_4 FAD-oxidase_C FAD-oxidase_C Fer4_8 CCG CCG DUF3400 PGPGW

JAHKAE010000192.1:200-8538
Proteobacteria bacterium

BrnT_toxin ? Unstab_antitox ParE_toxin HATPase_c_3 DNA_mis_repair MutL_C IPT IPPT TPRs tRNA-synt_1 Anticodon_1 Val_tRNA-synt_C

JAHIIA010000327.1:1544-11633
Proteobacteria bacterium

BrnA_antitoxin BrnT_toxin MoeA_N MoCF_biosynth MoeA_C TRCF Helicase_C DEAD CarD_TRCF_RID UvrB_inter Rotamase_2 SurA_N_3 Rotamase_3 SurA_N ABC_tran UvrA_DNA-bind UvrA_inter ABC_tran

JAHIKV010000221.1:1-8212
Planctomycetota bacterium

? ? ? BrnT_toxin Prismane PSP1 ? CBM9_1 PSII_BNR PSII_BNR PSII_BNR ?

JAGWZO010000360.1:1-3376
Nitrospirae bacterium

COX1 ? Phos_pyr_kin BrnT_toxin ?

JABARR010000094.1:9063-19829
Fimbriimonadaceae bacterium

UDG ? PEP-CTERM Sigma70_r2 Sigma70_r4_2 ? ? Sortilin-Vps10 BrnT_toxin ? ? DUF4172 Fic ? ? Phage_holin_3_6 ?

JAFAUJ010001068.1:1-3177
Hyphomicrobiales bacterium

FAD_binding_4 FAD-oxidase_C HTH_3 BrnT_toxin ?

JAFAVK010000138.1:8291-21281
Acidobacteriota bacterium

Peptidase_M2 baeRF_family10 eRF1_3 Zn_Ribbon_1 Acetylase ADH_N ADH_zinc_N ADH_zinc_N_2 BPL_LplA_LipB BPL_C TonB_C BrnT_toxin ? NUDIX Anthrone_oxy ? Glyco_transf_28 EryCIII-like_C Pan_kinase DUF5995

JAFDKU010000308.1:284-5766
Deltaproteobacteria bacterium

Calcineurin DUF2196 Spermine_synth DUF188 BrnT_toxin BrnA_antitoxin ? Acyl-CoA_dh_1 Acyl-CoA_dh_M Acyl-CoA_dh_N ?

JAHHHT010000004.1:102031-115026
Oscillatoria princeps RMCB-10

? PIN VapB_antitoxin ? NMT1_2 Uma2 CBM_20 Lipase_GDSL_2 DUF1796 BrnT_toxin ? ? BrnT_toxin ? AMP-binding AMP-binding_C BPD_transp_1 ABC_tran

JAIBBE010000057.1:38393-11276
Fimbriimonadaceae bacterium

Complex1_51K SLBB_2 NADH_4Fe-4S Receiver GerE ? Cys_Met_Meta_PP ? BrnT_toxin NadA ? FAA_hydrolase_N FAA_hydrolase ClpB_D2-small AAA_2 AAA PEP-CTERM ? ?

JAIBBV010000012.1:15438-28532

Oligoflexia bacterium

? ? DNA_processg_A DprA_WH ? ? Na_H_Exchanger TrkA_N HTH_1 LysR_substrate BrnT_toxin BrnA_antitoxin HIT ? ParE_toxin HTH_Tnp_1 TnpB_IS66 LZ_Tnp_IS66 zf-IS66 DDE_Tnp_IS66 DDE_Tnp_IS66_C

JAIESJ010000090.1:4198-9089
Burkholderiales bacterium

His-Kinase PAS 6TM_PilS ? CMD Y1_Tnp Phage_integrase BrnT_toxin ? polyprenyl_synt Ribosomal_L21p Ribosomal_L27 GTP1_OBG MMR_HSR1 AA_kinase PUA CNP1

JAIEPK010000664.1:1-8544
Candidatus Binatia bacterium

DUF222 ? CoaE ? BrnT_toxin ? DUF4062 pYEATS ? CoA_transf_3 PTE

JAIOJV010000118.1:36564-16757
Gammaproteobacteria bacterium

Uma2 DHquinase_II PEP-CTERM Toluene_X Big Big Big Poly_export SLBB BrnT_toxin BrnA_antitoxin MtrB_PioB BBP2_2 Wzz GNVR ParA AAA_22 VanZ Inositol_P

JAIOJA010000040.1:40180-53805
Planctomycetota bacterium

DUF1844 ? PDT ACT Aminotran_1_2 ? ? ATG22 BrnT_toxin ? Transket_pyr Transketolase_C ? NAD_Gly3P_dh_C NAD_Gly3P_dh_N G3P_acyltransf MTHFR S-methyl_trans polyprenyl_synt

JAIOHS010000243.1:10-5670
Hyphomicrobiales bacterium

Dus Autotransporter DUF1289 gag-asp_proteas CobS DBI_PRT BrnT_toxin HTH_3

JAIVJK010000023.1:1-8225
Acidobacteria bacterium

SHMT BrnT_toxin ? DEAD DEAD_2 Helicase_C_2 MbtH PTPS ? ? HTH_8 AAA_lid_14 NtrC-AAA Receiver

JAIVIS010002315.1:349-1513
Actinomycetota bacterium

? ? BrnT_toxin

JADCIQ010000221.1:2565-3599
Rhodocyclaceae bacterium

BrnA_antitoxin BrnT_toxin Aldo_ket_red Fic_N Fic SoFic-like_C ? ?

JADBWR010000472.1:2175-10246
Pseudanabaena sp. M114S2SP2A07QC

ArsA_ATPase ArsA_HSP20 Big Uma2 ketoacyl-synt ketoacyl-synt Ketoacyl-synt_C PemK_toxin ? ? BrnT_toxin ?

JAGQKM010001905.1:1-1055
Candidatus Omnitrophota bacterium

Lyase_1 BrnA_antitoxin BrnT_toxin

JAGRDC010000296.1:239-4820
Anaerolineales bacterium

BrnA_antitoxin BrnT_toxin ABC1 Peptidase_A8 Glutaredoxin ? ? DUF1028

JAGRCQ010000901.1:47-2368
Caldilineaceae bacterium

TIM ? BrnT_toxin ? PGK

JAGRCM010001410.1:1-1641
Caldilineaceae bacterium

DNA_pol3_beta_3 BrnT_toxin ? HIT Peptidase_M24

JAGQXF010000767.1:1879-2153
Saprospiraceae bacterium

BrnT_toxin ? HxlR DJ-1_PfpI ?

JAGRIQ010000081.1:1137-7633
Pseudomonadales bacterium

DksA-like_N ? PrmA BrnA_antitoxin BrnT_toxin Epimerase BrnA_antitoxin BrnT_toxin LysR_substrate HTH_1 DEAD RecG_wedge

JACKBU010000011.1:11515-14078
Ardenticatenaceae bacterium

? AA_kinase Aminotran_3 ? BrnT_toxin Peptidase_M20 HCS_D2 HMGL-like Aconitase Aconitase Aconitase_C Sod_Fe_N Sod_Fe_C Malic_M malic Malic_M malic GFO_IDH_MocA_C3 GFO_IDH_MocA

JACKDM010000015.1:4368-19985
Myxococcales bacterium

DNA_alkylation adh_short AlkZ-like LNS2_PITM1-3 PITM1-3 ? TrAA12 Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 BrnT_toxin BrnA_antitoxin ? HAD DUF1304 Peptidase_M15_4 Receiver STY-kinase

JACKIH010000003.1:3147452-3157763
Rhodospirillaceae bacterium

adh_short_C2 Ldh_2 DctP DctQ DctM Aldo_ket_red SIS_2 BrnT_toxin BrnA_antitoxin Hydrolase_6 Hydrolase_like PEMT Fructosamin_kin ? FUR ?

JADLEL010000054.1:12874-28745
Pirellulales bacterium

? STY-kinase TatA_B_E ? ? Sigma70_r4_2 Sigma70_r2 SQHop_cyclase_N DUF4159 DUF4159 BrnT_toxin ? FHA ? Trigger_N Trigger_C CLP_protease CLP_protease ?

JADLBU010000038.1:23306-40986
Phycisphaerales bacterium

tRNA-synt_2c tRNA_SAD DHHA1 ? N_methyl ? Transket_pyr Transketolase_C DUF433 DUF5693 BrnT_toxin E1_dh Discoidin Glyco_hydro127C Glyco_hydro127M Beta-AFase-like_GH127_cat BPD_transp_1 BPD_transp_1 CoA_trans ? Zn_ribbon_TFIIB Zn_ribbon_TFIIB Ribosomal_S21

JADYXZ010000004.1:161667-8449
Bradyrhizobiaceae bacterium

Sigma70_r3 Sigma70_r2 Sigma70_ner Sigma70_r1_2 Sigma70_r1_1 ? DnaB_bind Toprim_4 DNAG_N Zn_ribbon_DnaG BrnT_toxin BrnA_antitoxin Acyl-CoA_dh_N Acyl-CoA_dh_2 HTH_20 AHSA1 Glyoxalase Tautomerase YqeY

JADZDN010000152.1:150-10424
Alphaproteobacteria bacterium

Urease_alpha Amidohydro_1 Amidohydro_1 MatE MatE MatE MatE BrnT_toxin BrnA_antitoxin Epimerase Thiolase_N Thiolase_C_1 AMP-binding AMP-binding_C PCC PCC adh_short_C2

JADYZI010000317.1:1-10156
Acidobacteria bacterium

Ni_insertion GTP_cyclohydroI PA Peptidase_M28 PhdYeFM_antitox PIN tRNA-synt_1g tRNA-synt_1g Anticodon_1 tRNA_bind BrnT_toxin PsbP_2 TatD_DNase DUF520 polyprenyl_synt

JAJQBO010000128.1:107-7437
Planctomycetaceae bacterium

BrnA_antitoxin BrnT_toxin ? ? ? S6OS1 MobA_MobL YdaS_toxin YdaS_toxin OLD-like_TOPRIM AAA_15

JAJONL010000128.1:1-7515
Nitrospira sp.

? Ribonuc_red_2_N LAGLIDADG_3 Ribonuc_red_lgC Ribonuc_red_lgC TSCPD DUF4359 BrnA_antitoxin BrnT_toxin HTH_Tnp_1

JAIPFL010000042.1:12871-19899
Chromatiaceae bacterium

Rhodanese Rhodanese ? ? ? GGDEF ? LysE BrnT_toxin ? Helicase_C SNF2-rel_dom

JAKLPV010000122.1:5500-22122
Anaerolineae bacterium

MoCF_biosynth Peptidase_M32 Aminotran_1_2 ? ? Aminoglyc_resit BrnA_antitoxin BrnT_toxin Aminotran_1_2 ? DUF4325 His-Kinase ? Thioredoxin EamA EamA GFO_IDH_MocA_C3 GFO_IDH_MocA

JAKSDM010001039.1:1-3452
Pseudanabaenales cyanobacterium

? PemK_toxin Phage_scaffold BrnT_toxin ? GUN4 CopG_antitoxin BrnT_toxin

JAKSBU010000391.1:1-1356
Pirellulales bacterium

? BrnT_toxin ? Hydantoinase_B

JAGFCA010000172.1:3413-13169
Candidatus Poribacteria bacterium

CarboxypepD_reg CarboxypepD_reg CarboxypepD_reg CarboxypepD_reg CarboxypepD_reg ? ? ? Rhodanese MauE Sulfotransfer_5 BrnT_toxin ? PSII_BNR Sortilin-Vps10 Sortilin-Vps10 DUF6242

JADFKK010000022.1:28819-16543
Planctomycetota bacterium

Peptidase_S51 AhpC-TSA PSCyt2 PSD1 PSD1 BrnT_toxin ? DUF1501 Beta-prop_THOC3 Beta-prop_EML Peptidase_M50 DUF6576 HD_5 Uma2 Peptidase_S41 PDZ_6

JADFHR010000070.1:4422-12904
Proteobacteria bacterium

Alginate_exp N6_Mtase Methylase_S Transposase_20 DEDD_Tnp_IS110 MqsA_antitoxin BrnT_toxin FTSW_RODA_SPOVE PBP_dimer Transpeptidase MreD MreC MreB_Mbl HMGL-like HCS_D2 LeuA_dimer TPRs TPRs

JADFHH010000183.1:1396-5039
Proteobacteria bacterium

PCMT DHO_dh DUF952 CSD Iso_dh DOPA_dioxygen BrnT_toxin ?

JAKEHG010000720.1:68-13770
Gemmataceae bacterium

DUF1501 ? DUF433 ? ? BrnT_toxin PSCyt1 PSCyt2 PSD1 RecG_dom3_C Helicase_C DEAD RecG_wedge BD-FAE ? LpxB Sulfotransfer_3 ?

JAKEGE010000387.1:138-5042
Acidobacteria bacterium

BrnA_antitoxin BrnT_toxin UxuA UxuA GFO_IDH_MocA GFO_IDH_MocA_C2 GFO_IDH_MocA_C GFO_IDH_MocA_C2 DAO GMC_oxred_N GMC_oxred_C

JAKEEU010000323.1:72053-86930
candidate division KSB1 bacterium

Poly_export SLBB ? STAS HTH_21 Glyco_transf_5 Glycos_transf_1 GalP_UDP_transf GalP_UDP_tr_C Fib_alpha T7SS_EccA1_N ? BrnT_toxin OMP_b-brl FTSW_RODA_SPOVE FTSW_RODA_SPOVE PBP_dimer Transpeptidase MreD MreC MreB_Mbl MGS AICARFT_IMPCHas

JALESJ010000073.1:148016-165226
Bifidobacterium animalis

Rrf2 NAD_binding_10 Pyr_redox_2 Hydrolase E1-E2_ATPase PDZ_2 Trypsin Hydrolase TGT BrnT_toxin CopG_antitoxin Peptidase_M48 ? SRAP MFS_1 NAD_binding_10 Peripla_BP_3 LacI

JAGMBN010000124.1:626-3119
bacterium

Gln-synt_C Y1_Tnp ? ? BrnT_toxin

JAGLIB010000755.1:1-5015
Desulfobacterales bacterium

KAP_NTPase ? BrnT_toxin FRG ? ? ?

JAKLJP010000009.1:123643-149142
Anaerolineae bacterium

? NUDIX FtsK_4TM FtsK_alpha FtsK_SpoIIIE FtsK_gamma Uma2 HhH-GPD Ada_Zn_binding HTH_18 Methyltransf_1N DNA_binding_1 ? BrnT_toxin Clp_N Clp_N AAA AAA_lid_9 AAA_2 ClpB_D2-small FtsX MacB_PCD FtsX MacB_PCD ? ? MutL ? ?

WQWL01000090.1:8961-10152
Oscillospiraceae bacterium

SWIM DUF6429 MarR_2 YOMG_N BrnT_toxin Resolvase Recombinase Zn_ribbon_recom Bro-N HTH_3 HAD_2 Sigma70_r4_2 Cys_rich_VLP ?

WRFA01000120.1:2481-4655
Micrococcales bacterium

Gly_kinase ? HTH_23 BrnT_toxin BrnA_antitoxin 2H ? Acetylase

WQXO01000003.1:54801-67228
Oscillospiraceae bacterium

? YcxB ? Acetate_kinase HIGH_NTase1 HIGH_NTase1 Sigma70_r2 Sigma70_r4_2 ? BrnT_toxin GPI Sigma70_r2 Sigma70_r4_2 ? Methyltransf_11 ? SMI1_KNR4 ?

WQXD01000042.1:11703-19169
Oscillospiraceae bacterium

UPF0029 ? POR Fer4 TPP_enzyme_C POR_N Transketolase_C Transket_pyr Transketolase_N ? BrnT_toxin GPI

JAHDWJ010000018.1:41498-53304
Anaerolineae bacterium

HTH_31 AAA_22 T2SSF TPRs TPRs TPRs TPRs ? SdrD_B GmrSD_N SLFN_AlbA_2 ? BrnT_toxin Peptidase_C39_2 ? HEPN NTP_transf_2 Abhydrolase_1 Receiver GerE HAMP HisKA_3 His-Kinase

JAMLHW010000005.1:168224-183971
Anaerolineae bacterium

BPD_transp_1 BPD_transp_1 ECH_1 Ferritin ? DUF1801 Cupin_2 ? BrnT_toxin tRNA-synt_1 tRNA-synt_1 Anticodon_1 DUF5915 Mrr_N Mrr_cat DUF3253 ? tRNA-synt_2c tRNA_SAD ABC1 NMO

JAAELT010000358.1:312-10267
bacterium

? RNaseH-CATRA+Caspase TPR_MalT TPRs TPRs ? ? TPRs ? BrnT_toxin

JAAENG010000344.1:3764-17966
Chloroflexota bacterium

? SWIM ? DPPIV_N Beta-prop_THOC3 PD40 ? VitK2_biosynth ? BrnT_toxin Radical_SAM CofH_C PMT_2 UbiA ? DUF5678 ? ?

JAKSSG010000013.1:29503-11324
Clostridia bacterium

Lys-AminoMut_A OAM_dimer B12-binding CoA_trans CoA_trans RUBY_RBDX Rubrerythrin BrnT_toxin CopG_antitoxin Arginase Amidohydro_3 RMMBL Lactamase_B_2 Peptidase_M3 Peripla_BP_6 NifU

JAIKWK010000128.1:5329-13231
Bacillota bacterium

GGDEF ? P-loop_TraG PrgI ? TrbL_2 BrnA_antitoxin BrnT_toxin TrbC T4SS-DNA_transf ?

JAMXFE010000062.1:7408-17030
Laspinema palackyanum D2b

Ribosomal_L35p VTT_dom Glycos_transf_1 Glyco_transf_4 ? ? ABC_membrane_2 ABC_tran ? BrnT_toxin ? ? ?

JALOOB010000135.1:264-7834
Anaerolineae bacterium

BrnT_toxin ? PFO_beta_C TPP_enzyme_C POR_N POR FCD GntR Glyco_transf_4 Glycos_transf_1 AhpC-TSA NTP_transf_5

JALOPH010000056.1:1385-15616
Desulfobacterales bacterium

GTP_EFTU GTP_EFTU_D2 EFG_III EFG_IV EFG_C ? DUF4124 SLT tRNA-synt_2 Hist_deacetyl HicB_lk_antitox ? BrnT_toxin Mrr_N Hydant_A_N Hydantoinase_A dCMP_cyt_deam_1 GlnD_UR_UTase GlnE GlnD_UR_UTase GlnE PilZ P-II

JALORU010000167.1:4343-10289
Rubritepida sp.

DJ-1_PfpI ? Rce1-like eCIS_core PG_binding_1 Muramidase LysE BrnA_antitoxin BrnT_toxin

JAHCIP010000144.1:6966-14346
Anaerolineae bacterium

Rieske SufBD_N SUFBD_core NifU_N Aminotran_5 Uma2 NifU_N ? BrnT_toxin Rhodanese Rhodanese SufBD_N SUFBD_core

JAHCIO010000081.1:2657-12093
Anaerolineae bacterium

Glyco_trans_1_4 Glyco_trans_4_4 DUF711 ? DDE_3 SpoU_methylase MRM3-like_sub_bind MG2 Peptidase_M23 PGK BrnT_toxin ? Methyltransf_11

JAPHSS010000020.1:173848-187106
Gammaproteobacteria bacterium

HATPase_c_4 SLFN_AlbA_2 GmrSD_N GmrSD_C GmrSD_N Methylase_S Methylase_S N6_Mtase HsdM_N HTH_3 NLPC_P60 LRAT BrnT_toxin RPAP1_N ? ? HTH_31 ? ? ?

JAPLGS010000075.1:50481-55407
Chloroflexota bacterium

? DUF6036 HepT-like NTP_transf_2 Cupin_2 MarR_2 BrnA_antitoxin BrnT_toxin NERD

JAPLGM010000016.1:14968-25896
Campylobacterales bacterium

Fer4_7 Nudt16-like Acetylase ? ? Y1_Tnp ? BrnT_toxin Sigma70_r2 Sigma70_r4 ? zf_CopZ HATPase_c_3 HSP90 HSP90 PseudoU_synth_1 PseudoU_synth_1 LptF_LptG A24_N_bact Peptidase_A24

JAPLCU010000051.1:56-6206
Armatimonadota bacterium

? BrnT_toxin ? TPRs TPRs TPRs ? Alpha_L_fucos LTXXQ LTXXQ LTXXQ Metal_resist LTXXQ

JAPKYS010000021.1:22225-35375
bacterium

zf-dskA_traR DUF5915 Anticodon_1 tRNA-synt_1 DUF5890 PNPase DivIVA YGGT Ala_racemase_N BrnT_toxin ? SDF ? ? Aldedh EamA EamA CMD

JAPLSS010000226.1:1-7052
Candidatus Sumerlaeota bacterium

? BrnT_toxin ? ? UPF0150 HicA_toxin Hydrolase_3 S6PP ? Glycos_transf_2

JAPLRZ010000055.1:71453-92524
Methylococcales bacterium

PRiA4_ORF3 ATP-grasp_3 MCH Thiolase_N Thiolase_C 3HCDH_N 3HCDH ECH_1 Band_7_1 GYF ? BrnT_toxin ? HTH_8 AAA_lid_14 NtrC-AAA Receiver His-Kinase His-Kinase OEP OEP HlyD_D23 ACR_tran YecR

JAPLNJ010000959.1:4286-6502
Planctomycetota bacterium

HZS_alpha ? BrnT_toxin ?

JAPLOD010000478.1:1-10027
Planctomycetota bacterium

PA14 PQQ_2 PQQ_2 URO-D ? BrnT_toxin DUF2950 Hydrolase_like DUF1638 ? DUF3969 ThiC_Rad_SAM

JAPOOQ010000052.1:1-4899
Gemmatimonadetes bacterium

PIN VapB_antitoxin AdoHcyase ? HSP20 BrnA_antitoxin BrnT_toxin BcrAD_BadFG

JAPOOJ010000562.1:126-8060
Bryobacterales bacterium

? DUF4160 DUF2442 ? PD40 DPPIV_N Peptidase_S9 BrnA_antitoxin BrnT_toxin HTH_36 HTH_Tnp_1 HTH_21 rve rve_3 Pps Pps

JAOSKM010000001.1:753512-770050
Planctomycetota bacterium

Voltage_CLC CBS CBS Voltage_CLC Lactamase_B Trypsin PDZ_2 Nsp1_C PDZ_2 Dus ? BrnT_toxin MlaE ABC_tran MlaD Cytochrom_C_asm GlutR_dimer Shikimate_DH GlutR_N His-Kinase His-Kinase GARS_C GARS_A GARS_N

JAQCGV010000085.1:2965-7120
Proteobacteria bacterium

SpoU_sub_bind HTH_12 OB_RNB CSD2 RNB S1 ? BrnT_toxin ? DUF6904 PIN VapB_antitoxin ? CcdB CcdA

JAPTWG010000069.1:3183-24612
Nitrospiraceae bacterium

N_methyl N_methyl GspH ? ? His-Kinase His-Kinase HAMP Receiver HTH_8 AAA_lid_14 NtrC-AAA Receiver BrnT_toxin Alginate_exp ACR_tran ACR_tran HlyD_D23 OEP OEP YtkA Trans_reg_C Receiver His-Kinase His-Kinase HAMP

JAOJOV010000036.1:1-4581
bacterium

HTH_32 PRiA4_ORF3 BrnT_toxin ? HTH_3 HTH_3 ZnR STY-kinase

JAQGMG010000010.1:9045-19393
Rhodocyclales bacterium

Abhydrolase_6 DUF1232 AlkZ-like Spermine_synth DUF7910 Glyco_hydro_19 LIAS_N Radical_SAM ? BrnT_toxin LplA-B_cat DUF493 Glyco_hydro_46 Glyco_hydro_46 PEP-CTERM

JAQTMZ010000003.1:137580-151946
Gallionella sp.

Dala_Dala_lig_N Dala_Dala_lig_N Dala_Dala_lig_C POTRA_1 FtsQ_DivIB_C SHS2_FTSA FtsA Tubulin FtsZ_C LpxC ABC_tran ABC_tran_Xtn ABC_tran ? BrnT_toxin ? ? ? DUF2157 Methyltransf_10 KTSC Peptidase_M20

JAQURM010000004.1:41221-8573
Desulfovibrionales bacterium

HRDC HTH_40 BrnT_toxin CopG_antitoxin TPRs Unstab_antitox DUF4136 ? ? Radical_SAM

JAQUOA010000004.1:10743-25358
Bacteroidota bacterium

CsgG ABC_tran TonB_dep_Rec_b-barrel Plug ABC2_membrane_4 MFS_1 MFS_1 LolA_like ? BrnT_toxin CPSase_sm_chain GATase DapB_C DapB_N DHDPS Hexapep_2 THDPS_N_2 Receiver NtrC-AAA AAA_lid_14 HTH_8 Beta-prop_WDR5 WD40_Prp19 DMT_6

JAPPNG010000150.1:24733-39066
Spirochaetaceae bacterium

AP_endonuc_2 rhaM ? AI-2E_transport Peptidase_M29 HNH_5 Pribosyltran BrnT_toxin BrnA_antitoxin ? ABC-ATPase ? SBP_bac_5 ? ?

JAPPMR010000536.1:1-3814
Candidatus Poribacteria bacterium

? DUF7310 DUF932 BrnT_toxin FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 UnbV_ASPIC

JAPPEF010000255.1:4452-20819
Gemmatimonadota bacterium

? BCA_ABC_TP_C ABC_tran Sigma54_DBD Sigma54_CBD Sigma54_AID ? LptE Metallopep Reprolysin_5 Beta-prop_THOC3 Beta-prop_THOC3 Big FlgD_ig ? BrnT_toxin Metallopep Reprolysin_5 Beta-prop_THOC3 Beta-prop_THOC3 Big FlgD_ig Lar_N TPP_enzyme_C TPP_enzyme_N ? DUF2064 NYN_YacP

JARFXP010000010.1:1901-15383
Nitrospira sp.

CBS CBS SIS CDP-OH_P_transf AXE1 G3P_acyltransf ? ? ? BrnT_toxin ? MutL ? DUF6754 ? ? ?

JAOTXM010000380.1:1-4059
Candidatus Tectomicrobia bacterium

Band-7 Guanylate_cyc cNMPBD TIR BrnT_toxin BrnA_antitoxin

JAOTXV010001317.1:45-2117
Alphaproteobacteria bacterium

SBP_bac_8 MqsA_antitoxin BrnT_toxin ?

JAOUAU010000197.1:534-3218
Gammaproteobacteria bacterium

BrnT_toxin BrnA_antitoxin DUF4347

JASEGY010000040.1:3200-14510
bacterium

HypD ? ? DUF3417 Phosphorylase Phosphorylase DUF433 DUF34_NIF3 BrnA_antitoxin BrnT_toxin His_biosynth Glucosaminidase LysM LysM LysM Asp_protease_2 PdxJ SIS SIS GATase_6

JASJDR010000287.1:130-8015

Xenococcus sp. MO_188.B8

? ? BrnT_toxin ? ? Phycobilisome LplA-B_cat Na_H_Exchanger Apolipoprotein Receiver AMP-binding AMP-binding_C

JAKDMT010000354.1:49-2462
Nitrosospira sp.

ParE_toxin ? BrnT_toxin ? HicA_toxin ?

JAUNEQ010000255.1:1-2195
Coriobacteriia bacterium

Enolase_C BrnT_toxin CopG_antitoxin CbiQ

JAUSDQ010000324.1:1-6494
Sulfuricaulis sp.

PrmA ? ? ? HNH_repeat HNH_repeat HNH ? BrnT_toxin DNA_topoisoIV DNA_gyraseA_C DNA_gyraseA_C

JAUSOZ010000012.1:15964-35348
Deltaproteobacteria bacterium

ABC2_membrane_2 ABC_transp_aux DUF7088 DUF4340 DUF4340 SurE Fer2_4 Fer4_7 Fer4_20 Pyr_redox_2 Fer4_20 2Fe-2S_thioredx Complex1_51K NADH_4Fe-4S Fer4_21 BrnA_antitoxin BrnT_toxin FTHFS DUF5981 MTHFR Prismane DUF4191 Thioredoxin Pyr_redox_2

JAUXMS010000796.1:1-2900
Nitrospirota bacterium

He_PIG BrnT_toxin ? ? TumE ATP-synt_E ? DUF2283

JAUYFD010000530.1:1-1616
bacterium

? Oxidored_q2 BrnT_toxin ? Oxidored_q4

JAUYCI010000096.1:268-3817
Anaerolineales bacterium

Robl_LC7 V4R Glycos_transf_2 Glycos_transf_2 NTP_transf_2 HepT-like ? BrnT_toxin

JALYHZ010000126.1:1-4628
Pseudomonadota bacterium

Glyoxalase AHSA1 3-dmu-9_3-mt ABC_tran ABC_tran_Xtn ABC_tran BrnT_toxin HTH_3 BrnA_antitoxin Hydrolase Cpn60_TCP1

JALYVY010000178.1:9582-9410
Gemmatimonadota bacterium

? ? ? ? DDE_Tnp_1_6 BrnT_toxin BrnA_antitoxin MazE_antitoxin PemK_toxin

JALYZF010000253.1:8522-23649
Acidobacteriota bacterium

Peptidase_M28 Abhydrolase_11 4HBT PseudoU_synth_1 PseudoU_synth_1 Fer2 Sod_Fe_N Sod_Fe_C LIAS_N Radical_SAM BrnT_toxin ? Acetylase Acetylase Peptidase_M16 Peptidase_M16_C Peptidase_M16 Peptidase_M16_C PAP2 SpoIIM ABC_membrane ABC_tran

JAUZRX010000017.1:141082-154927
Acidobacteriota bacterium

AAA_lid_2 AAA_3 Spermine_synth Spermine_synt_N AdoMet_dc ? ? Peptidase_S8 Peptidase_S8 BrnT_toxin CopG_antitoxin TSP_3 TSP_3 FG-GAP FG-GAP FG-GAP FG-GAP FG-GAP ? ? SEC-C Xre_MbcA_ParS_C EPSP_synthase

JAISFK010000055.1:409-12074
Clostridiales bacterium

? BrnT_toxin Cas_Cas5d ? Cas_Cas7 Cas3-like_C_2 DEAD HD Bgal_small_N LacZ_4 Glyco_hydro_2_C LamininG Glyco_hydro_2_C Glyco_hydro_2 Glyco_hydro_2_N

JAITIE010000131.1:4142-12877
Candidatus Margulisiibacteriota bacterium

ABC_tran FtsX MacB_PCD ? ? ? Trns_repr_metal YgjP-like YgjP-like BrnT_toxin BrnA_antitoxin PALP Cys_Met_Meta_PP Rrf2

JARLGA010000049.1:16063-12097
Terracidiphilus sp.

TonB_C HTH_31 HigB-like_toxin HTH_3 DnaJ DnaJ_C BrnT_toxin ? MerR_1 Glyco_transf_21 GT87 PMT_2 MarC Sirtuin TIR

JAVLCL010000182.1:1-6653
Halothece sp. Uz-M2-17

Methylase_S PIN ? ? BrnT_toxin HsdM_N N6_Mtase SocA_Panacea ? ?

JAVQWA010000388.1:359-2443
Roseiflexaceae bacterium

PIN ? BrnT_toxin ? Amino_oxidase

JANRLS010000198.1:12410-26467
Pirellulaceae bacterium

TPRs TPRs tRNA_bind Anticodon_3 tRNA-synt_1g ? ? Calcineurin Glyco_trans_1_3 DUF1501 BrnT_toxin ? ? UPF0004 Radical_SAM TRAM N6_Mtase

JANRJB010000073.1:1-7291
Rhodospirillaceae bacterium

PIN BrnA_antitoxin BrnT_toxin ? AhpC-TSA BNR_2 ? NAD_binding_10 Ribonuc_L-PSP Flavin_Reduct PCP_red

JAXHTF010000018.1:32167-60047
Acidobacteriota bacterium

? PEMT ? LamininG SGSH_C Sulfatase ABC_tran ABC_membrane MacB_PCD FtsX MacB_PCD FtsX Carbam_trans_N Carbam_trans_C BrnT_toxin BrnA_antitoxin Thioesterase PP-binding AMP-binding_C AMP-binding Condensation PP-binding AMP-binding_C AMP-binding Condensation Condensation PP-binding AMP-binding_C AMP-binding PP-binding AMP-binding_C AMP-binding Condensation PP-binding AMP-binding_C AMP-binding Condensation

JAXMJK010000098.1:4524-6325
Gemmatimonadales bacterium

Transketolase_C NAD_kinase NAD_kinase_C SMC_N ? ? BrnT_toxin ?

JAXMJV010001056.1:1-5860
Candidatus Binatia bacterium

FA_desaturase MFS_1 MFS_1 GST_N_2 GST_C ? BrnT_toxin ? HicB_lk_antitox HicA_toxin ? DUF2442

JAWXBA010000400.1:1324-3269
Gemmatimonadota bacterium

? BrnT_toxin ? BrnA_antitoxin PemK_toxin RHH_1 PhdYeFM_antitox ParE_toxin

JAXHNI010000009.1:277470-287363
Gammaproteobacteria bacterium

TPMT Acetylase DMT_6 DUF6946 PIN7 DUF2442 ? BrnT_toxin DEDD_Tnp_IS110 Mrr_N Mrr_cat ? DUF2283 ? ? AAA_lid_2 AAA_3

JAYEHB010000617.1:605-3430
Thermodesulfobacteriota bacterium

? BrnT_toxin ? Uma2 MTTB

JAYWMR010000403.1:362-1894
Nitrospirota bacterium

BrnT_toxin BrnA_antitoxin CbiQ

JAYZKT010000034.1:82-6027
Pseudomonadota bacterium

BrnT_toxin TPRs BPD_transp_2 BPD_transp_2 Peripla_BP_6 ?

WHSP01000160.1:1-9750
Gemmatimonas sp.

FtsX MacB_PCD FtsX MacB_PCD Acetylase Dala_Dala_lig_N Dala_Dala_lig_N Dala_Dala_lig_C ? BrnT_toxin DUF932 NikA-like Xre_MbcA_ParS_C HTH_23 AAA_33 HTH_17 Phage_int_SAM_5

SHVQ01000047.1:1214-12378
Rhodospirillales bacterium

K_trans K_trans_C Ribonuc_L-PSP Ribonuc_L-PSP Peptidase_M20 2-Hacid_dh_C CoA_transf_3 Urocanase_N Urocanase Urocanase_C Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 BrnT_toxin HTH_3 DpaA_N 2-Hacid_dh_C

SHYO01000091.1:641-10290
Chloroflexi bacterium

Beta-lactamase ? BrnT_toxin Methyltransf_11 Trm112p zf-C6H2 DUF1003 ? MOSC HhH-GPD ? PriA_3primeBD DEAD Helicase_C PriA_C

VXOF01000017.1:4449-15560

Nitrospira sp. SB0662_bin_26

TraY DS Thioredoxin_4 Aldedh PvlArgDC Synaptojanin BrnA_antitoxin BrnT_toxin ? DUF2726 AnmK ? Zip Zip

VYFE01000106.1:1-6250
Rhodospirillaceae bacterium

Hexapep Hexapep Ribosomal_L13 Ribosomal_S9 ? BrnT_toxin HTH_3 Semialdhyde_dh Semialdhyde_dhC_1 SnoaL_4 Methyltransf_25 Methyltransf_25

RGQH01000245.1:1-4640
Proteobacteria bacterium

CTP_transf_3 BrnT_toxin ? UPF0175 ? DUF3368 Glyco_tran_28_C Formyl_trans_N NeuB

JAAGRQ010000042.1:25411-32773
Desulfolutivibrio sulfodismutans

BPD_transp_1 ? AAA_2 ClpB_D2-small Colicin_V MazG MazG HEPN_RiboL-PSP HTH_3 BrnT_toxin Helicase_C DEAD

JAAHFM010000184.1:160-9371
Symploca sp. SIO1A3

? DNA_methylase BrnT_toxin BrnA_antitoxin ? STY-kinase FHA DUF1232 MotA_ExbB Thioesterase Hydrolase

JAAHFM010000249.1:815-9639
Symploca sp. SIO1A3

GDT1 PhdYeFM_antitox ParE_toxin CxxCxxCC Mg_chelatase AAA_lid_2 VWA_2 DUF6932 HTH_3 BrnT_toxin BrnA_antitoxin ? Avidin ? STY-kinase

PHGD01000002.1:16978-30714
Nitrospira sp. LK70

DUF5338 NikA-like Relaxase TraI_middle LPD7 TraI-like_C T4SS-DNA_transf T4SS-DNA_transf Peptidase_S26 HTH_3 HTH_37 BrnT_toxin BrnA_antitoxin TraI_middle ? TraI-like_C T4SS-DNA_transf Peptidase_S26 SDF

JAACAH010000770.1:3153-16490
candidate division KSB1 bacterium

Polbeta ? His-Kinase HAMP His-Kinase ? Glyco_transf_4 Glycos_transf_1 BrnA_antitoxin BrnT_toxin PIG-L DUF1841 Cupin_2 ? ? Unstab_antitox Methyltransf_25

JAAUTN010000689.1:1593-1629

Leptolyngbyaceae cyanobacterium SM1_4_3

DUF4096 DUF6887 ? BrnT_toxin TPR_NPHP3

JAAUUS010000002.1:4710-19363

Hydrococcus sp. SU_1_0

DUF928 ? RNaseH-CATRA+Caspase Tic22 Caspase DUF4384 Phage_int_SAM_1 Phage_integrase BrnT_toxin ? ? ? ? ? ? ?

JAAUPR010000055.1:2971-16382

Limnothrix sp. RL_2_0

SMC_N ABC-ATPase DUF4276 PutA_N Pro_dh Aldedh BrnT_toxin ? AAA_19 UvrD_C ? FAA_hydrolase Rv2993c-like_N Ribosomal_S6 TOBE_2 OB_MalK ABC_tran TPRs TPRs ChAPs

JABFGE010000003.1:262453-272780
Planctomycetota bacterium

DDE_Tnp_1 ? HTH_28 ? RelB-like ? BrnA_antitoxin BrnT_toxin ? IstB_IS21 HTH_23 rve Mu-transpos_C_2 ? ? ? ?

JABFSC010000383.1:1-5856
Deltaproteobacteria bacterium

DUF4158 DDE_Tnp_Tn3 RPA HTH_3 BrnT_toxin

JAADII010000120.1:38408-44249
Chloroflexota bacterium

Methyltransf_11 HicA_toxin UPF0150 DegT_DnrJ_EryC1 Epimerase Glycos_transf_2 ? BrnT_toxin Sulfotransfer_1

MBRE01000078.1:106377-119709
Oscillatoriales cyanobacterium USR001

CAF1A_acidic HIT Photo_RC ? Pribosyltran ? Guanylate_cyc Receiver His-Kinase His-Kinase GAF GAF Receiver BrnT_toxin ? REase HEATs HEAT_EZ HEATs HEATs HEATs NACHT NCH4

MEKI01000099.1:1681-8627

Acidobacteria bacterium RBG_13_68_16

? BrnA_antitoxin BrnT_toxin Methyltransf_25 SNase HTH_Tnp_1 ? bPH_2 Transgly_assoc ? DoxX

MFSU01000082.1:1-11931

Candidatus Muproteobacteria bacterium RBG_16_65_34

UvrD_C UvrD_C UvrD-helicase PDDEXK_1 BrnA_antitoxin HTH_8 Receiver His-Kinase His-Kinase 6TM_PilS Plug TonB_dep_Rec_b-barrel ? BrnT_toxin

MGBV01000222.1:8114-27822

Candidatus Rokubacteria bacterium GWC2_70_16

DctP HGSNAT_cat ? FAA_hydrolase NIPSNAP NIPSNAP GFO_IDH_MocA GFO_IDH_MocA_C3 TPRs TPRs TPRs TPRs BrnT_toxin adh_short_C2 CMD Creatininase Creatininase BPD_transp_1_N BPD_transp_1 Amidase SBP_bac_5 ADH_N ADH_zinc_N OppC_N BPD_transp_1

MGSG01000237.1:11010-34416

Deltaproteobacteria bacterium RIFCSPLOWO2_02_FULL_53_8

MreB_Mbl ? MreC PBP_dimer Transpeptidase PgpA RecX_HTH1 RecX_HTH2 RecX_HTH3 BrnT_toxin DUF6978 DUF1828 DUF1829 FeoB_N FeoB_Cyto Gate FeoB_C Gate Fe_dep_repr_C FeoA LexA_DNA_bind Peptidase_S24 T2SSE CopG_antitoxin FUR

MGUN01000009.1:14010-27927

Elusimicrobia bacterium RIFCSPHIGHO2_02_FULL_39_36

Carb_kinase Peptidase_M20 AIRS AIRS_C TsaE TsaD Aminotran_4 4HBT BrnT_toxin ? PDZ_6 TPRs YfiO TPRs TPRs YfiO YfiO YfiO ARM_TT21_C MotA_ExbB ExbD TonB_C ExbD

MHEU01000027.1:24878-4964

Nitrospirae bacterium RIFCSPLOWO2_02_FULL_62_14

MoCF_biosynth 2Fe-2S_thioredx AIG2_2 DUF4124 ? Rieske HNH_5 BrnT_toxin

MNCT01000080.1:6030-8219

Nitrospirae bacterium 13_2_20CM_62_7

BrnA_antitoxin BrnT_toxin NUDIX GCV_T GCV_T_C FIST FIST_C ? ? Lysine_decarbox Peptidase_S58

MQVI01000062.1:1-11486
Pasteurellaceae bacterium 15-036681

? Methylase_S Methylase_S Bro-N HsdM_N N6_Mtase BrnT_toxin TGS tRNA_SAD tRNA-synt_2b HGTP_anticodon GST_N_3 Glutaredoxin2_C Glycos_transf_2 Glycos_transf_2 ? ? CopG_antitoxin Maf

NVSG01000002.1:277010-295506
Hyphomicrobiales bacterium

ABC_tran OB_MalK TOBE_2 SIS HTH_6 HAD_2 MatE MatE BrxA WYL BrnT_toxin T4SS-DNA_transf Phage_integrase Phage_int_SAM_1 MR_MLE_C MR_MLE_N ? AMP-binding Prenyltrans Prenyltrans Prenyltrans PP-binding HTH_3

PHCA01000111.1:20909-34706
Chloroflexi bacterium HGW-Chloroflexi-1

DUF4013 Yip1 ABC_tran CARDB NPCBM_assoc NPCBM_assoc FtsX MacB_PCD Zn_ribbon_8 BrnA_antitoxin BrnT_toxin FTHFS Peptidase_C25 Propeptide_C25 PPC DUF11 DUF11 Peptidase_C25

QHTC01000174.1:176-3583
Candidatus Rokuibacteriota bacterium

RHH_1 PIN BrnT_toxin ? ? PDDEXK_1 ?

QHUJ01000029.1:164-3316
Gemmatimonadetes bacterium

? ? ? ? ? BrnT_toxin Phage_head_fibr

QHVJ01000317.1:4070-5596
Acidobacteria bacterium

tRNA-synt_2 N-OB HGTP_anticodon tRNA-synt_His DCTN5 Hexapep BrnT_toxin ? ?

QHVW01000293.1:5199-9229
Acidobacteria bacterium

HTH_8 AAA_lid_14 NtrC-AAA His-Kinase His-Kinase PAS HAMP ? BrnT_toxin ? Peptidase_M23 DUF3368 ? LANC_like ?

QHZK01000036.1:321-7007
Acidobacteria bacterium

BrnT_toxin ? TPRs ? Citrate_synt DUF3592 YqeY OPT

QHBM01000114.1:13249-23965
Chthoniobacterales bacterium

HATPase_c_3 ? ? PIN MazE_antitoxin HTH_37 ? Nramp ? ? ? FMN_dh PNPase BrnT_toxin Aldo_ket_red

CP036352.1:2876403-2898397
Phycisphaerae bacterium RAS2

PLDc_N HKD HKD PEP-CTERM Methyltransf_23 AMP-binding AMP-binding_C ? Ephrin_rec_like Ephrin_rec_like Ephrin_rec_like Ephrin_rec_like Ephrin_rec_like Ephrin_rec_like Ephrin_rec_like Ephrin_rec_like Ephrin_rec_like FAA_hydrolase BrnT_toxin ? Acetylase GreA_GreB GreA_GreB_N ? MraZ MraZ Glyco_trans_1_4 Glyco_transf_4 ?

CP064976.1:2001463-2017096
Deltaproteobacteria bacterium

CarboxypepD_reg MM_CoA_mutase Glu_synthase TatC TatA_B_E Hist_deacetyl CopG_antitoxin BrnT_toxin Peptidase_S8 DUF3047 ? DEAD Helicase_C DbpA ? ? Transgly Transpeptidase

QEUS01000232.1:1-4128
Anaerolineae bacterium

DUF6391 ? BrnT_toxin OsmC SdrD_B LysM NDH2_C Pyr_redox_2

QZKN01000183.1:28-14690
Candidatus Omnitrophota bacterium

AP_endonuc_2 Peripla_BP_4 KduI TAL_FSA Asp-Al_Ex TrkA_C TrkA_C Asp-Al_Ex BrnA_antitoxin Bmp ABC_tran ABC_tran BPD_transp_2 BPD_transp_2 Paired_CXXCH_1 Cytochrome_C554 ? GFO_IDH_MocA_C3 GFO_IDH_MocA BrnT_toxin

PYIY01000004.1:17785-31924
Candidatus Poribacteria bacterium

Glyoxalase PhyH PhyH ? DUF6784 DUF6785 MR_MLE_C MR_MLE_N BrnA_antitoxin Rce1-like MR_MLE_N MR_MLE_C ? CN_hydrolase Amidohydro_1 NUDIX PhyH BrnT_toxin

QNEL01000407.1:4371-7176
Candidatus Parabeggiatoa sp. nov. 3

Glyco_trans_2_3 Epimerase BrnT_toxin BrnA_antitoxin BrnT_toxin ? Glycos_transf_2 ?

RFIW01000941.1:11027-10152
Chloroflexi bacterium

AP_endonuc_2 NAD_binding_2 6PGD FGGY_N FGGY_C ? BrnT_toxin Uma2 PhdYeFM_antitox ParE_toxin PNPase Lar_N adh_short UxaE

RPPI01000194.1:1-8220
Chloroflexi bacterium

DUF4062 AP-ATPase DUF2442 ? BrnT_toxin ? ? CBS CBS B12-binding Radical_SAM DUF4070 PhoU_div PHO4 Metallophos_3

RPOY01000975.1:223-3460
Acidobacteria bacterium

? BrnT_toxin ? HEATs ?

RPOT01001109.1:1755-1505
Acidobacteria bacterium

? Glyoxalase_6 ? BrnT_toxin

RSAS01000855.1:363-8604
Candidatus Viridilinea halotolerans

PUCC ? BrnT_toxin ? PcrA_UvrD_tudor UvrD_C UvrD-helicase PDDEXK_3 Methyltransf_25 ?

RSAS01000596.1:10771-17790
Candidatus Viridilinea halotolerans

? ? Uma2 NTP_transf_2 HEPN AAA_22 ? BrnT_toxin ?

QNZO01000081.1:155-3108
Gammaproteobacteria bacterium

SufE SprT-like BrnT_toxin BrnA_antitoxin GIY-YIG Glucosamine_iso G6PD_C DUF2283

RDXH01000150.1:10609-7686
Oscillatoriales cyanobacterium

DUF2358 BrnT_toxin CopG_antitoxin DUF7689 ArsA_ATPase ArsA_HSP20 ? 2OG-FeII_Oxy_3 ANKs ANKs ANKs ANKs ANKs ANKs ?

RDXV01000002.1:25707-45806
Nostocales cyanobacterium

Beta-Roll Beta-Roll Beta-Roll Beta-Roll Beta-Roll FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 Rotamase ABC_tran ABC_membrane Peptidase_C39 cNMPBD HlyD_3 Biotin_lipoyl_2 DXPR_C DXP_redisom_C DXP_reductoisom ? BrnT_toxin DUF4007 ? ? ? BrxL_ATPase BrxL_N ? 4HBT COP23

SHND01000039.1:41468-62462
Chloroflexota bacterium

PD40 PD40 PD40 DUF6788 FMN_dh RNaseH-CATRA+Caspase Sirtuin nSTAND1 Beta-prop_WDR5 Beta-prop_WDR3_1st Beta-prop_THOC3 ? AAA BrnT_toxin ? Peptidase_C25 ? Guanylate_cyc Guanylate_cyc AAA_16 TPRs TPRs TPRs TPR_MalT Ferritin Peptidase_S10 ?

SMVQ01000342.1:126-6603
Planctomycetota bacterium

HEATs Peptidase_M23 ? BrnT_toxin ? ? Y2_Tnp Zn_Ribbon_1 TfoX_N Sigma70_ECF STY-kinase

SOHZ01000645.1:6214-19964
Anaerolineales bacterium

MFS_1 Ppx-GppA_III NAD_binding_2 NAD_binding_11 GAF_2 GAF GAF_3 Guanylate_cyc Band-7 Acetyltransf_9 ? BrnT_toxin CopG_antitoxin Abhydrolase_1 adh_short_C2 Glyco_tran_28_C MGDG_synth MFS_1 DUF951 Ribosomal_S6

SOHZ01000014.1:342-5155
Anaerolineales bacterium

BrnT_toxin BrnA_antitoxin Amidohydro_1 UPF0047 Radical_SAM_N2 Radical_SAM

VBOO01000163.1:4413-5801
Nitrospirae bacterium

? BrnT_toxin Fic MarR_2 Transglut_core2 HEATs Lactamase_B_2 MafB19-deam Bac_globin ? Transposase_mut

VBOO01000059.1:3124-13474
Nitrospirae bacterium

GerE ABC_tran UvrA_inter AAA_29 ABC_tran ORF6N Gp49 HTH_31 DUF4309 BrnA_antitoxin BrnT_toxin Ubie_methyltran Radical_SAM PCP_red RUBY_RBDX PCP_red Usp Usp

VBLE01000036.1:231-4513
Deltaproteobacteria bacterium

BrnT_toxin BrnA_antitoxin Amidohydro_2 DUF6883 adh_short_C2 adh_short adh_short DUF4926 DUF2281

VBDH01000055.1:1-6067
Betaproteobacteria bacterium

FAD_binding_3 ACT_4 TGS RelA_SpoT HD_4 EamA EamA Peptidase_M90 HNH_5 BrnT_toxin BrnA_antitoxin GFA

VBAB01000574.1:2096-5337
Alphaproteobacteria bacterium

Acetylase ? BrnT_toxin NAD_binding_2 NAD_binding_11 Aminotran_5 HTH_3

CP073041.1:5372638-5389298

Woronichinia naegeliana WA131

DDE_5 DDE_Tnp_1 DDE_Tnp_1_assoc HTH_32 DDE_3 Uma2 Uma2 Pyr_redox_3 DAO FAD-depend_C BrnT_toxin ? ? RuBisCO_small RbcX RuBisCO_large RuBisCO_large_N LplA-B_cat Ribosom_S30AE_C Ribosomal_S30AE

CAACVI010000004.1:25204-37938

Candidatus Desulfarcum epimagneticum

YidD 60KD_IMP YidC_periplas Gly-zipper_Omp TerB DnaJ DEAD Helicase_C DSHCT CbiA DUF2442 CopG_antitoxin BrnT_toxin ? ? Y1_Tnp Y1_Tnp Phage_integrase Y2_Tnp Zn_Tnp_IS91 Trypsin ?

CAADFE010000010.1:32974-38288
Candidatus Kentrum sp. FW

HicA_toxin ? DUF2283 fn3 YafQ_toxin ? ? ? BrnT_toxin ? ? ? DUF2283 DUF4258 ? ?

CP129466.1:616341-625539
Gammaproteobacteria bacterium

PRAI Acetyltransf_7 ParE_toxin ? EbsA ? VapB_antitoxin BrnT_toxin ARG HTH_3 Gp49 CopG_antitoxin BrnT_toxin Mrr_N Mrr_cat Acetylase

CP129483.1:1997541-2008808
Anaerolineales bacterium

SMC_N zf-B_box NAD_kinase_C NAD_kinase Cas_Cas4 DUF2157 Lactamase_B_2 cNMPBD HTH_Crp_2 BrnT_toxin ? 4HBT_2 Peptidase_M42 Bac_luciferase ? MazG HhH-GPD

CP129483.1:2362379-2377114
Anaerolineales bacterium

Frankia_peptide BTAD TPRs TPRs AAA_22 Inositol_P Receiver Arginase GGDEF PAS HisKA_7TM Amidohydro_1 Amidohydro_3 HUTI_composite_bact BrnT_toxin ? ? BD-FAE SCP2 DOT1 MFS_1 tRNA_edit

NZ_JADBSL010000039.1:8508-27183
Microcystis sp. M042S1

dCMP_cyt_deam_1 PspA PDGLE CbiQ Glyco_hydro_57 ? CbiM DUF1822 Peripla_BP_6 TPRs RNaseH-CATRA+Caspase Peripla_BP_6 TPRs Trans_reg_C Receiver DNA_gyraseB_C Toprim Intein_splicing DNA_gyraseB His-Kinase BrnT_toxin ABC_tran HEATs HEATs HEATs HEATs

NZ_JABWTO010000009.1:56026-16770
Photobacterium damselae subsp. damselae

YihI MMR_HSR1 Vut_1 PrlF_antitoxin 5_3_exonuc_N 5_3_exonuc DNA_pol_A_exo1 DNA_pol_A G_glu_transpept DEAD Helicase_C RecQ_Zn_bind RQC HRDC DUF3630 ? BrnT_toxin BrnA_antitoxin Cytochrom_C Cytochrom_C Zn_ribbon_IS1595 Methyltransf_23 Toxin_YhaV DUF2489

NZ_CP004387.1:3556773-3574491
Isoalcanivorax pacificus W11-5

AAA_lid_14 NtrC-AAA Receiver His-Kinase His-Kinase PAS ? zf-dskA_traR DksA_N Aminotran_1_2 SfsA_N SfsA BrnT_toxin BrnA_antitoxin Peripla_BP_6 tRNA-synt_2 Elong-fact-P_C EFP EFP_N ? PAS GGDEF EAL Radical_SAM tRNA-synt_1c tRNA-synt_1c

NZ_CAUVAK010000018.1:13005-30363
uncultured Selenomonas sp.

BrnA_antitoxin MlaE ? Sigma70_r2 DUF748 TamB TamB OEP OEP Flg_bb_rod LlgE_F_G_D1 FlgE_D2 Flg_bbr_C ECH_1 BrnT_toxin MlaD ABC_tran Cauli_VI RNase_H PagL Flavoprotein Seryl_tRNA_N tRNA-synt_2b

NZ_CP075171.1:3346054-3362428
Leptodesmis sichuanensis A121

CRISPR_Cas2 ? Cas_Cas1 Cas_Cas4 Uma2 Cas_Cas5d ? Cas8b_C Cas8b_N BrnT_toxin ? DUF433 Cas3_HD Cas6 WCX WYL HTH_11 Calcineurin GSDH

NZ_CP007002.1:1-3361
Campylobacter fetus subsp. venerealis cfvi03/293

MobC NikA-like Relaxase CopG_antitoxin BrnT_toxin Rep3_N Rep3_C

NZ_AP039366.1:1058537-1077595
Desulfovibrio sp. GTC20076

FapA Glyco_transf_9 BrnT_toxin Phage_AlpA BrnA_antitoxin Arm-DNA-bind_3 Phage_int_M Phage_integrase FlgD FLgD_tudor HTH_17 DUF3800 ? ? ? Flg_bb_rod LlgE_F_G_D1 FlgE_D2 Flg_bbr_C Flg_hook DUF3880 Glyco_trans_1_2

NC_016588.1:190380-207070
Azospirillum lipoferum 4B

RepA_C Asp_Arg_Hydrox PNPase HTH_OrfB_IS605 OrfB_IS605 Cas12f1-like_TNB MannoseP_isomer ManC_GMP_beta-helix NTP_transferase NTP_transferase BrnT_toxin ? ABC_tran ABC_membrane Peptidase_C39 HlyD_3 Glycos_transf_2 Epimerase BrnA_antitoxin GtrA_DPMS_TM ? ?

NZ_JAMJFP010000033.1:40227-48172
Methylobacter sp.

? Gp49 BrnT_toxin PTP-NADK ? GDE_N_bis MGH1-like_GH HTH_37 ? Cation_efflux ZT_dimer DUF4112

NZ_AUZJ01000034.1:96871-114822
Treponema socranskii subsp. socranskii VPI DR56BR1116 = ATCC 35536

SBP_bac_8 PMI_typeI_cat MPI_cupin_dom BrnA_antitoxin HD_assoc HD_assoc HD Guanylate_cyc AAA_16 Tyr_Deacylase Alpha-amylase CBM_48 PUD Bmp BrnT_toxin NikA-like MscL BPD_transp_1 BPD_transp_1 ACT_6 CitMHS ThiF

NZ_AQPY01000244.1:1646-5642
Microcystis aeruginosa PCC 7005

Ham1p_like BrnT_toxin BrnA_antitoxin ISP_coupler N6_Mtase Type_ISP_C Radical_SAM LIAS_N ?

NZ_JACFQK010000005.1:100135-111713
Snodgrassella sp. M0118

BrnA_antitoxin DUF4491 HTH_3 P63C TMP_3 ? ? SocA_Panacea BrnT_toxin CHAP Bactofilin Bactofilin ? ? HTH_3 SHOCT

NZ_JNJA01000006.1:845-13291
Solidesulfovibrio alcoholivorans DSM 5433

? BrnT_toxin Arm-DNA-bind_3 Phage_integrase TonB_C ? HTH_18 BrnA_antitoxin ExbD ? MotA_ExbB HTH_3 MFS_1 ? ? ?

NZ_JACJTA010000005.1:32900-46697
Scytonema hofmannii FACHB-248

ARG HD 7TM-7TMR_HD 7TMR-HDED Glycos_transf_1 Glyco_transf_4 Glycos_transf_1 Glyco_transf_4 BrnT_toxin ? ? AdoHcyase_NAD AdoHcyase MS_channel_1st MS_channel_2nd MS_channel_3rd DHquinase_II DMP19 ? ? ? HTH_3

NZ_FKIO01000139.1:1-10260
Candidatus Entotheonella palauensis

DHDPS Methyltransf_1N DNA_binding_1 DUF1326 BrnA_antitoxin BrnT_toxin Mob_synth_C Radical_SAM AI-2E_transport Sigma70_r4 Sigma70_r3 PKD HODM_asu-like PpoB2 Molydop_binding Molybdopterin Molybdop_Fe4S4 ?

NZ_CAMZOL010000011.1:92877-110647
Halomonas sp. Alg239-R46

Epimerase Peripla_BP_2 FecCD ABC_tran CbiA GATase_3 Nuc_deoxyri_tr3 CobA_CobO_BtuR BrnT_toxin ? nSTAND_NTPase5 ATP-synt_ab Helicase_C DEAD MerR-DNA-bind MerR_1 Glyoxalase BrnA_antitoxin

NZ_LJOQ01000022.1:23634-35451
Anabaena sp. AL09

? GFO_IDH_MocA_C GFO_IDH_MocA TOBE_2 ? PBP_dimer Transpeptidase HEATs HEATs HEATs DUF3458 Peptidase_M1 Peptidase_M1_N DUF3352 BrnT_toxin

NZ_LIRE01000243.1:4518-17090
Pseudanabaena sp. 'Roaring Creek'

? AAA_lid_3 AAA DUF3038 Uma2 PhdYeFM_antitox Unstab_antitox ? BrnT_toxin DUF7753 DUF433 KAP_NTPase ? DUF4143 AAA_14 SRP_SPB SRP54 SRP54_N

NZ_CP060204.1:213245-229616
Selenomonas timonae

BrnA_antitoxin DUF748 TamB TamB TamB OEP OEP Flg_bb_rod LlgE_F_G_D1 FlgE_D2 Flg_bbr_C RHH_1 ECH_1 MlaD ABC_tran MlaE Cauli_VI RNase_H PagL Flavoprotein Seryl_tRNA_N tRNA-synt_2b Flagellar_put BrnT_toxin

NZ_CP182900.1:15825-15739
Cyanobacterium sp. IPPAS B-1200

? ? ? MgtE Sigma70_r4 Dynamin_N ? Sigma70_r4 ? BrnT_toxin Phage_integrase Phage_int_SAM_1 DNA_pol_A_exo1 DNA_pol_A ? AEP-TOTE DnaB_C AEP-TOTE DnaB_C

NZ_LT828550.1:39984-51635
Desulfamplus magnetovallimortis

FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 PutA_N ParE_like LamininG HEPN NTP_transf_2 ? HicB HicA_toxin HicB_lk_antitox HTH_8 AAA_lid_14 NtrC-AAA Receiver Zn_Tnp_IS91 His-Kinase His-Kinase HAMP ? BrnT_toxin

NZ_CP021983.2:2229860-2245071
Halomicronema hongdechloris C2206

? WCX WYL HTH_11 Cas6 ? BrnT_toxin Cas8b_C Cas8b_N Cas_Cas5d ? Cas_Cas1 Cas_Cas4 CRISPR_Cas2 SecA_DEAD P-loop_SecA SecA_SW Acetyltransf_7 ? ? Cas3-like_C_2 DEAD Cas3_HD

NZ_AQQS01000028.1:51626-68824
Aurantimonas sp. 22II-16-19i

ABC_tran ABC_membrane Peptidase_M19 YiaAB PTA_PTB Malic_M malic BrnT_toxin BrnA_antitoxin DAO SoxD Fer2_4 Pyr_redox_2 SO_alpha_A3 GCV_T GCV_T_C SoxG FAD_binding_6 NAD_binding_1 Fer2 Gp_dh_C Gp_dh_N Rieske Ring_hydroxyl_A TetR_C_33 TetR_N PspA

NZ_FNNZ01000001.1:302368-13780
Thiocapsa roseopersicina

DUF2442 ? BrnT_toxin PIN Thioredoxin_2 MazE_antitoxin ? DUF1275 DUF4160 VanZ MmeI_N MmeI_hel MmeI_Mtase MmeI_TRD HTH_29 HTH_33 DDE_3

NZ_BMGN01000001.1:57272-75999
Niveispirillum cyanobacteriorum

META MliC YscW EAL Alpha-amylase Malt_amylase_C BrnT_toxin BrnA_antitoxin Aminotran_1_2 GntR NTP_transferase dTDP_sugar_isom Trp_halogenase RmlD_sub_bind GDP_Man_Dehyd ? ? Plug DUF983

NZ_PXOH01000088.1:125-2299
Aphanothece hegewaldii CCALA 016

DUF6364 BrnT_toxin ? ? ?

NZ_QNRK01000014.1:130143-143860
Roseiarcus fermentans

Acetylase TacA1 ? Acetylase ? adh_short AMP-binding AMP-binding_C HTH_3 Methyltransf_11 HK Lactamase_B DUF2019 RNase_T BrnT_toxin Pro_CA

NZ_CP030126.1:2354306-2366814
Indioceanicola profundi

DUF4398 OmpA DUF4398 Fasciclin PRC ? CoA_binding Ligase_CoA ATP-grasp_2 Ligase_CoA Patatin Ldh_1_N Ldh_1_C ? PepSY_2 Acetylase BrnT_toxin BON BrnA_antitoxin

NZ_RDQK01000003.1:269-19882
Allofranklinella schreckenbergeri

VWA_CoxE MoxR-AAA Acetylase 2-oxogl_dehyd_N E1_dh Transket_pyr OxoGdeHyase_C Biotin_lipoyl E3_binding 2-oxoacid_dh BrnT_toxin Pyr_redox_2 Pyr_redox_dim LytTR Receiver ? ? IPTL-CTERM Flg_new_2 Flg_new_2 Flg_new_2 Flg_new_2 Flg_new_2 Flg_new_2 Flg_new_2 DUF5122 DUF5122 DUF5122 ? IPTL-CTERM His-Kinase His_kinase Peptidase_M23

NZ_SPNT01000017.1:8247-24351
Nitratireductor sp. XY-223

? PhdYeFM_antitox HTH_18 AraC_binding_2 CN_hydrolase PD40 PD40 PD40 BrnT_toxin HTH_3 HTH_23 HTH_32 HTH_33 DDE_3 SNF2-rel_dom Helicase_C T4SS-DNA_transf TraD TraC ? DUF4096 DDE_Tnp_1 PIN

NZ_VDUZ01000048.1:25636-35945
Vineibacter terrae

HTH_20 RNaseH-CATRA+Caspase FHA Pepco DUF2442 ? BrnT_toxin PemK_toxin MazE_antitoxin DUF2282 MbnB_TglH_ChrH AHSA1 ? NTP_transf_3 4HBT_3C 4HBT_3 HTH_3

NZ_CP045423.1:2928766-2945336
Microvirga thermotolerans

SDF Catalase-rel Catalase HTH_1 LysR_substrate COesterase MgtC DUF4010 Sigma70_r4 Sigma70_r3 Sigma70_r2 Sigma70_ner Sigma70_r1_2 Sigma70_r1_1 DnaB_bind Toprim_2 DNAG_N Zn_ribbon_DnaG BrnT_toxin BrnA_antitoxin Isochorismatase YqeY CPSase_sm_chain GATase ? Methyltransf_25 AsnC_trans_reg HTH_AsnC-type

NZ_JACIGK010000022.1:20869-37689
Roseospira visakhapatnamensis

TonB_dep_Rec_b-barrel Plug TP_methylase TP_methylase CbiC TP_methylase TP_methylase CbiG_N BrnA_antitoxin BrnT_toxin CbiJ ? FAD_binding_2 Flg_bbr_C LlgE_F_G_D1 Flg_bb_rod ABC1 Ubie_methyltran Radical_SAM HemN_C

NZ_JACJSD010000056.1:2656-13606
Nostoc sp. FACHB-280

? ? ? ATP-synt_DE_N ATP-synt_DE PIN BrnT_toxin BrnA_antitoxin TPRs TPRs TPRs TPRs Phage_holin_6_1 MNS-STAND HTH_26 MNS-STAND ? Band-7 ? ? ? ?

NZ_CAWPPI010000134.1:122748-16611
Iningainema tapete BLCC-T55

Radical_SAM ABC_tran ABC_membrane AAA_16 ? BrnT_toxin ? ? Glutaredoxin GSH-S_N GSH-S_ATP FtsZ_C Tubulin FtsQ_DivIB_C POTRA_1 GAF SpoIIE GTP-bdg_N GTP-bdg_M MMR_HSR1 MMR_HSR1_Xtn ? RXYLT1_C

NZ_JACFXY010000005.1:146038-161052
Mixta mediterraneensis

DUF1525 TraU ? ? TraG_N ? BrnA_antitoxin BrnT_toxin HTH_Tnp_1 Sigma70_r4_2 HTH_21 rve rve_3 PhdYeFM_antitox HerA_C DUF445 LptF_LptG LptF_LptG rve_2 rve HTH_21 HTH_Tnp_1

NZ_JADEXH010000028.1:53079-64665
Sphaerospermopsis sp. LEGE 08334

polyprenyl_synt ? NUDIX AI-2E_transport ? ? BrnT_toxin STY-kinase TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs ABC-ATPase ? ? HigB_toxin RelB-like HTH_3 Peptidase_M78 THF_DHG_CYH THF_DHG_CYH_C

NZ_CP053482.1:3444779-3455061
Thiothrix subterranea

? ? CLP_protease Cast ? ? BrnA_antitoxin BrnT_toxin ProQ Phage_integrase Receiver GerE AAA-ATPase_like PDDEXK_9 UPF0175 PhdYeFM_antitox DUF3368 PIN-TPR-GreABC PilJ HAMP HisKA_3 His-Kinase

NZ_JAEKJR010000001.1:1132553-1143220
Microbulbifer salipaludis

NUDIX DUF1249 ? His-Kinase DNA_gyraseB Toprim DNA_gyraseB_C ? BrnT_toxin ? ? DUF4864 DUF4440 ? ? ? Calcineurin UPF0227

NZ_JAGINP010000023.1:143161-157026
Azospirillum rugosum

CBS CBS TerC S4 ? HSP90 HATPase_c_3 BrnA_antitoxin BrnT_toxin ? Bact_transglu_N Transglutaminase FUR ? HAD DEXQc_Suv3 Helicase_C CBS His-Kinase His-Kinase

NZ_JAHBBI010000007.1:141395-165361
Bifidobacterium pluvialisilvae

FGGY_C FGGY_N ABC_tran Ala_racemase_C Ala_racemase_N DnaG_DnaB_bind DnaB_bind Toprim_4 DNAG_N Zn_ribbon_DnaG UDG CopG_antitoxin BrnT_toxin Helicase_C SNF2-rel_dom SNF2_assoc SWIM ? ? ? EamA EamA GmrSD_N GmrSD_C RAMA Abhydrolase_2 ?

NZ_JAIQZM010000040.1:757-19064
Argonema galeatum A003/A1

PG_binding_1 MJ1316 Synaptojanin 2H PAP_NTPase PolyA-Pol HD AAA_33 Beta-prop_THOC3 Beta-prop_WDR3_1st STY-kinase Uma2 ? BrnT_toxin Oxidored_molyb TPRs TPRs TPRs TPRs TPRs TPRs TPRs Pps Pps Pps Methyltransf_21 PurK_C ATP-grasp RS_preATP-grasp-like Peptidase_M16_C Peptidase_M16 Peptidase_M16_C Peptidase_M16 RNA_ligase

NZ_CANMAK010000017.1:20108-34013
Sulfitobacter mediterraneus

? ? ? GmrSD_N GmrSD_C HTH_3 HTH_19 BrnA_antitoxin BrnT_toxin PutA_1st ParE_toxin WGR ATPgrasp_TupA TPRs TPRs TPRs Sulfotransfer_3 DUF4214 DUF4214 RPA

NZ_CATNLY010000006.1:49095-20503
Pleomorphomonas sp. T1.2MG-36

Terminase_2 ? DrHyd ? MFS_1 MFS_1 BrnT_toxin BrnA_antitoxin ? EAL GGDEF CHASE2 FecR HAMP MCPsignal ABC_tran ABC_tran

NZ_CADBFY010000005.1:17742-30824
uncultured Selenomonas sp.

? ? STY-kinase HTH_3 FBPase_2 ? BrnT_toxin DUF5672 Glyco_tranf_2_4 YdjM DUF2939 PTS_EIIA_1 ? Abi_2 ? PBECR4

NZ_WFYD01000062.1:42112-57844
Sulfurivirga sp.

Pterin_4a Peptidase_M48_N Peptidase_M48 RNase_T TPP_enzyme_C TPP_enzyme_M TPP_enzyme_N ALS_ss_C ACT_AHAS_ss HTH_31 BrnT_toxin VTT_dom Hydrolase Hydrolase E1-E2_ATPase Cation_ATPase_N PEP-utilizers_C PEP-utilizers PPDK_N MFS_1 MFS_1 HTH_5 ArsP_1 CMD KARI_C KARI_N

NZ_JALEZZ010000024.1:38884-52380
Selenomonas sp.

CheX MFS_1 HSP20 ? Glyco_trans_1_4 Glycos_transf_2 BrnA_antitoxin BrnT_toxin Aminotran_3 ALAD TP_methylase HEM4 Porphobil_deam Porphobil_deamC GlutR_N Shikimate_DH GlutR_dimer

NZ_CAXAZZ010000001.1:135066-29957
uncultured Pseudomonas sp.

bMG10 CUB_A2MG TED_complement A2M A2M_BRD bMG6 bMG5 MG2 bMG3 A2M_bMG2 bMG1 ? BiPBP_C Transpeptidase Transgly ShlB POTRA_3 POTRA_2 Fil_haemagg_2 Fil_haemagg_2 Fil_haemagg Fil_haemagg Fil_haemagg Fil_haemagg Fil_haemagg Fil_haemagg Fil_haemagg Fil_haemagg Fil_haemagg Fil_haemagg Fil_haemagg Fil_haemagg Fil_haemagg Fil_haemagg Fil_haemagg Fil_haemagg Fil_haemagg TPS rve rve_3 BrnA_antitoxin BrnT_toxin DUF2247 ? DUF637 Fil_haemagg_2 Fil_haemagg_2

NZ_DAIDSI010000016.1:11205-21281
Acidiferrobacter sp.

Phosphonate-bd XFP_C XFP XFP_N SBP_bac_11 PhdYeFM_antitox HTH_3 ? BrnT_toxin BrnA_antitoxin PIN

NZ_CAKZKO010000171.1:4026-11613
uncultured Paraglaciecola sp.

? ? ? ? ? ? BrnA_antitoxin BrnT_toxin VirC1 ? RPA RelE RelE HTH_3 ?

NZ_CANLQB010000022.1:676-9207
uncultured Roseobacter sp.

? ? Relaxase NikA-like T4SS-DNA_transf T4SS-DNA_transf HTH_17 BrnA_antitoxin BrnT_toxin ? RPA

NZ_JAUXNS010000033.1:11818-29793
Parvibaculum sp.

Glyco_transf_28 Glyco_tran_28_C Mur_ligase Mur_ligase_M Mur_ligase_C FAD_binding_4 MurB_C Dala_Dala_lig_N Dala_Dala_lig_N Dala_Dala_lig_C POTRA_1 FtsQ_DivIB_C SHS2_FTSA FtsA OEP OEP HlyD_D23 ACR_tran VitK2_biosynth Methyltransf_21 KAP_NTPase ? BrnT_toxin Tubulin FtsZ_C

NZ_DAOREB010000071.1:1-12702
Plasticicumulans sp.

MCPsignal HAMP HAMP sCache_2 MeaB B12-binding EAL Receiver Hpt Receiver His-Kinase His-Kinase PAS PAS GAF_2 PAS PAS Methyltransf_11 ? ABM BrnT_toxin ?

CP159373.1:99357-116125
Candidatus Electrothrix aestuarii

PhdYeFM_antitox PIN ParE_toxin PhdYeFM_antitox PIN ? BrnT_toxin Macro ? DUF6531 RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat Tox-ART-HYD1 Acetylase SEC-C SecA_SW P-loop_SecA P-loop_SecA SecA_DEAD UPF0102 PilZ PemK_toxin